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BYD, China, and Global Electric Vehicle Rivalry

By any standard, BYD has come from nowhere to be one of the major car companies in the world. [. . .] What they've done in a short period of time is nothing less than stunning.

— — Former Tesla Director Steve Westly¹

In March 2023, Shenzhen, China-based electric vehicle (EV) maker BYD passed Tesla to become the world's top-selling electric vehicle maker by units,² delivering just over 548,000 passenger EVs globally.³ Tesla worldwide deliveries that quarter came to 423,000 (see **Exhibit 1**).^{4,a} China, the world's leading car exporter,⁵ was on a tear developing its domestic EV market, spending an estimated \$173 billion on incentives, tax exemptions, and EV infrastructure between 2009 and 2022. In the first quarter of 2023, one quarter of the nearly 1.1 million vehicles that China exported were electric.⁶ Analysts predicted that by 2025, 13% of Chinese passenger vehicles on the road would be electric, compared to 6% globally,⁷ and half of all vehicles sold in China would be EVs.⁸

Entrepreneur Wang Chuanfu founded BYD, short for Build Your Dreams, as a rechargeable mobile phone battery manufacturer in 1995. After buying an automobile manufacturer in 2003, and with an investment from Warren Buffet in 2008, Wang drove BYD hard into China's EV market. He said, "[A]s petrol was set to create an (environmental) [sic] problem, we needed to start developing electric cars. The future of electric cars in China is very bright."⁹ He was right: between mid-2021 and the start of 2023, BYD tripled its share of the global EV market, while others such as Volkswagen (VW) saw their share fall by as much as half (see **Exhibit 2**). Wang exhorted his China colleagues to "demolish the old legends and achieve new, world-class brands."¹⁰ The effort was nothing short of existential. He added, "[I]t's an emotional need for the 1.4 billion Chinese people to see a Chinese brand becoming global."¹¹

China, the world's largest car market,¹² was home to 94 domestic EV brands offering more than 300 models.¹³ One analyst described the Chinese market as "ultra-competitive" and "brutal," especially for foreign brands.¹⁴ EVs were a priority for Chinese investment, and Chinese brands controlled 81% of the domestic EV market.¹⁵ China's 2023 EV revenues were forecast to reach nearly \$300 billion, in stark contrast to U.S. EV revenues, projected to hit about \$70 billion.¹⁶ That did not stop U.S. and European

^a Figures for BYD included not just battery-powered EVs but also plug-in hybrid EVs, which included an auxiliary internal combustion engine (ICE). For more on EV variants, refer to **Appendix A**.

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carmakers from setting up beachheads in China to be close to Chinese buyers and parts suppliers. GM's CEO Mary Barra acknowledged the pressure: "China has 100 vehicle brands vying for sales and a 50% capacity utilization rate."¹⁷ A European Union (EU) official added, "[Western multinationals] have to be there. It keeps you fit."¹⁸

As foreign automakers scrambled to get a foothold in China, Chinese automakers competed globally with Ford, GM, Hyundai, Toyota, VW, and other "old legends."¹⁹ To win the global EV war, Chinese EV maker XPeng's vice chair said, "You cannot be a China-only player, you have to be a global player. [. . .] [I]n that scenario, maybe close to half your volume is coming from outside of China."²⁰ BYD had entered European markets – Norway in 2021, Sweden in 2022,²¹ and Germany in 2023.²²

Price competition in their home market pressured Chinese EV makers to sell overseas. In the first quarter of 2023, Tesla slashed prices to compete,^{b23} launching a race to the bottom that strained BYD and threatened smaller EV makers' viability.^{c24} Reporters dubbed China "ground zero for the price war in electric vehicles."²⁵ Between October 2022 and May 2023, Tesla lowered its prices another 20%;^{d26} a month later, Li Auto, NIO, and XPeng's EV sales were down a combined 60%.²⁷

One analyst warned that China's price wars were "unsustainable."²⁸ She added that the EV market might punish carmakers who slashed costs for short-term market share: "In the long-term, competition in the [EV] sector will come from areas such as vehicle intelligence, sales and service network expansion and infrastructure, which all require long-term capital input."²⁹

BYD already had multiple battery plants in China, and in July 2023, it began to extend its production offshore, announcing that it would invest \$624 million to support battery and EV production in a Brazilian plant purchased from Ford Motor Co. (Ford).³⁰ Earlier that year, Brazil had entered talks with Argentina, Chile, and Bolivia to control the supply of lithium, a key EV battery component.³¹ (See **Appendix A** for details about the technology and natural resources required to power EVs.)

Sometimes belatedly, incumbent automakers in Germany, Japan, South Korea, and the U.S. had all realized the need to field credible EVs and were racing to catch up. Many of these companies had global brands, over a century of automobile supply chain expertise, established dealer networks, and decades of experience developing and bringing new vehicles to market. Yet these "old legends" were often unable to match startups' agility. In its 20-year existence, Tesla had leapfrogged them all to become the world's most valuable automaker by market capitalization (see **Exhibit 3**).

Now, Chinese EVs were starting to crowd the stage; in the words of a former Tesla director, "Some would say they've out-Tesla'd Tesla."³² Ford CEO Jim Farley acknowledged, "The ones I'm paying the most attention to are Hyundai/Kia, the Chinese and Tesla. That's my list."³³ Could BYD maintain its lead? Were legacy automakers capable of competing with these upstart firms? If so, how?

BYD's Growth Trajectory

Born to a family of farmers, entrepreneur Wang Chuanfu in 1995 moved to Shenzhen and, at age 29, started BYD as a rechargeable lithium mobile-phone battery manufacturer with his cousin Lu

^b Tesla lowered its Model 3 sedan in China by 14% and its Model Y SUV by 10%.

^c One consultancy forecast that price competition could wipe out 140 of the 167 registered EV brands in China. Reuters, "Tesla Offers New China Rebate as Price Cuts Rock EV Market," July 7, 2023, <https://tinyurl.com/5a5u2xft>, accessed July 2023.

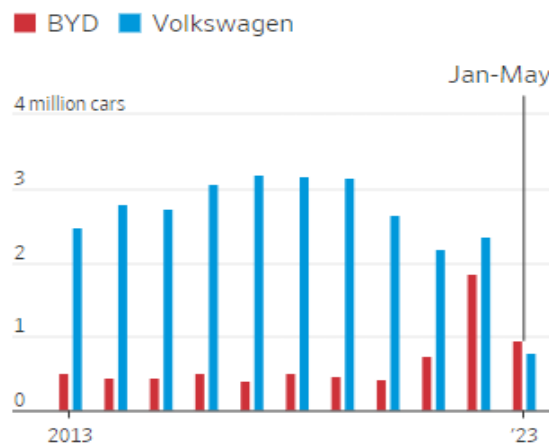
^d Tesla lowered the price of its Model Y an additional 20%.

Xiangyang.³⁴ Armed with his background in battery technology and his training as a chemist, Wang grew the enterprise quickly. In 2002, BYD acquired Qinchuan Automobile Company, an underperforming state-owned enterprise, including its license to sell cars in China.^e BYD began to manufacture its own semiconductors³⁵ and forward integrated from components into vehicle production,³⁶ presenting its first fully EV concept within two years.³⁷ In 2004, *Forbes* reported, “In the vast and booming Chinese auto market now dominated by foreigners, a small Chinese company called BYD Auto is barely noticeable [. . .] yet it would be foolhardy for GM, Volkswagen and other foreign makers to ignore [. . .] BYD.”³⁸

In 2008, Warren Buffet bought a 10% stake in BYD for \$230 million.³⁹ That December, the company sold China’s first homegrown mass-market EV, the F3DM,⁴⁰ and in 2010, made *Bloomberg*’s influential “Tech 100” list.⁴¹ In the 2010s, BYD continued growing at a rapid clip, and between 2016 and 2022, the company doubled its global workforce to 570,000 employees—200,000 more workers than were employed by Toyota, the world’s leading carmaker.⁴² In 2022, BYD turned heads by selling 1.86 million EVs—more vehicles than in the previous four years combined—recording \$60 billion in revenue.⁴³ In 2023, BYD expected to sell 3.7 million EVs, more than Tesla’s anticipated 2 million units for the year.⁴⁴

In the first quarter of 2023, BYD unseated VW as China’s best-selling car brand, a title VW had held since 2008 (see **Figure A**).⁴⁵ “BYD Is Racing Toward the Top of the Global EV Market,” one headline ran.⁴⁶ By July 2023, Buffet’s investment was worth \$8 billion,⁴⁷ BYD’s market cap was \$99.7 billion, and the company’s dealer network spanned 53 countries.⁴⁸ In 20 years, BYD had transformed itself from a maker of mobile phone batteries into one of the world’s leading automakers.

Figure A BYD and Volkswagen Sales in China, 2013-May 2023 (in units)



Source: Yoku Kubota and Selina Cheng, “In China, the Era of Western Carmakers Is Over,” *The Wall Street Journal*, July 10, 2023, <https://tinyurl.com/5beu8vm4>, accessed July 2023.

^e Qinchuan Automobile Company began as an artillery manufacturer, but by the 1990s manufactured a small car: the ALTO. For more on BYD’s acquisition of Qinchuan Automobile Company, refer to Robert S. Huckman and Alan D. MacCormack, “BYD Company Ltd.,” HBS No. 606-139 (Boston: Harvard Business Publishing, 2006).

Vertical Integration: More than the Sum of Parts

BYD manufactured its own EV batteries and semiconductors,^f making it the world's first fully vertically integrated EV maker. Battery cells and semiconductors were the two most expensive components in an EV.⁴⁹ Wang boasted to *Forbes*, “[We] mastered the core technologies of the whole industrial chain [of EVs], such as batteries, motors, and electric controls.”⁵⁰ (**Exhibit 4** shows information on EV battery component supply and demand.) One observer commented: “BYD is the most vertically integrated company that I’ve ever seen, not just in the auto space.”⁵¹

An auto industry expert found that “the high level of vertical integration in [BYD’s] battery supply chain gives it a clear cost advantage over similar carmakers, an enabler for rapid expansion both within and outside of China.”⁵² A Toyota researcher estimated that BYD’s vehicle production costs averaged 20% to 30% less than Toyota’s.⁵³ BYD passed the savings on to its customers, stimulating demand for its EVs. One investor predicted that BYD’s vertical integration would allow the company to sell more than 10 million vehicles per year by 2030.⁵⁴ “Other than windshields and tires,” he explained, “they can make on their own just about everything in the car. They have their own construction company that helps build factories. That’s how they can speed things up.”⁵⁵ In December 2022, BYD bought six ships to guarantee that EV shipments reached their destinations on time.⁵⁶

BYD was the world’s second-biggest manufacturer of EV batteries.⁵⁷ In 2021, it announced it would start selling EV batteries to other companies;⁵⁸ third-party demand accounted for 6.6% of BYD battery production that year.⁵⁹ In early 2023, BYD operated 11 battery manufacturing plants in China.⁶⁰ That February, the company announced that it was investing \$2.1 billion in two new battery factories in China’s Henan and Zhejiang provinces.⁶¹ Upon completion, BYD predicted that the two factories would produce batteries at a rate sufficient to power 600,000 new EVs every year.⁶²

BYD and other Chinese battery producers would combine to produce three times the battery capacity needed for domestic car-making by 2025, according to estimates.⁶³ An analyst observed, “Top players like BYD do not seem to be worried about overcapacity, as they continue to expand production. It is expected that more car assemblers will use BYD as their battery supplier.”⁶⁴ BYD’s signature EV battery, the Blade, appeared in BYD’s vehicles and in its competitors’—in October 2022, Toyota announced a small Blade-powered sedan developed jointly with BYD that would be produced and sold in China, and in 2023, Tesla started using the Blade in some of its vehicles.⁶⁵

In September 2022, BYD’s semiconductor unit opened a new production line in China’s Hunan Province that could manufacture 500,000 automotive chips every year,⁶⁶ approximately 5% of the annual production capacity of the world’s leading EV semiconductor manufacturer, Germany-based Infineon.⁶⁷ Two months later, a BYD filing mentioned that it planned to make several more “significant” investments in chip production capacity.⁶⁸ The unit’s output, the company wrote, needed to catch up to runaway demand for EVs.⁶⁹

BYD operated three major EV production bases in China—in Guangdong, Hunan, and Shaanxi provinces. In 2022, the company’s vehicle production capacity, which included a line of public buses, was 2.9 million units—and rising.⁷⁰ Tesla’s annual production capacity, by comparison, was 2 million units.⁷¹ Tesla did not make commercial vehicles, instead focusing on three sedans and a sports utility vehicle (SUV).⁷² By the end of 2023, Wang estimated that BYD’s annual production capacity would reach 4.3 million units.⁷³

^f In an EV, batteries replaced combustion engines and semiconductors enabled a range of in-demand vehicle technologies.

A Framework for EV Competition

Worldwide, EVs comprised 5% of new vehicle sales in 2020, 9% of new vehicle sales in 2021, and 14% in 2022.⁷⁴ The International Energy Agency expected EV sales to continue seeing “exponential growth.”⁷⁵ In 2023, one EV researcher proposed a three-part framework for analyzing EV competition:

EV acceptance depends on supply and demand. Three stakeholders directly influence this market: governments, manufacturers, and consumers—all of which are intrinsically linked. For example, a government push to improve charging infrastructure could lead to consumers growing more confident about driving range and charging availability, thus improving sales and acceptance. In this complex market, the impetus for EV acceptance cannot be placed on a single factor. Government goals, incentives, and legislation all influence the targets of original equipment manufacturers (OEMs), as seen with Volkswagen and the EU-27. A more varied market can lead to increased EV awareness, and, in turn, consumer purchasing intentions can impact the political discourse.⁷⁶

Manufacturers

Electric vehicles dated back to 1890, when a chemist in the U.S. state of Iowa first attached an electric battery to a six-seater wagon.⁷⁷ But market forces conspired to make gas-powered vehicles the dominant worldwide passenger vehicle type until 1997, when Toyota launched the Prius, the world’s first mass-market hybrid electric vehicle, in Japan.⁷⁸ (Refer to **Appendix A**.) In 2000, Toyota released the Prius worldwide, and the model became an overnight best-seller.⁷⁹ In 2006, Tesla Motors began production of its Roadster sports car, the first mass-produced battery EV that could travel more than 200 miles on a single charge.⁸⁰ The model’s success inspired legacy automakers to produce EVs of their own, including Nissan’s Leaf and GM’s Chevrolet Volt,⁸ which both hit the U.S. market in 2010.⁸¹

In the 2010s, as its economy recovered from the 2008 Global Financial Crisis, the Chinese government began investing in its domestic EV industry. Over the following decade, companies such as BYD, Wuling, Changan, and XPeng took a range of battery EVs and hybrids to market (see **Exhibit 5**). By 2022, EVs generated \$284.8 billion in revenue for Chinese companies, more revenue than EV sales generated for any other national economy.⁸² By then, the major players in the global EV industry included Tesla, Toyota, a generation of newer Chinese brands, and late-comer, legacy automakers vying for market share, ranging from BMW to Ford.

By 2021, EV product offerings varied widely from one manufacturer to the next. Toyota, for example, did not make a single battery EV, but produced 54,000 plug-in hybrids (for a comparison of hybrid electric and battery EVs, see **Exhibit 6**).⁸³ GM produced 13,000 battery EVs, but no plug-in hybrids.⁸⁴ (For a comparison of Chinese, East Asian, European, and U.S. EV manufacturers and their product offerings, see **Exhibits 7** and **8**). Time-to-charge and range also varied. In 2023, EV drivers could travel an average 220 miles on a single charge,⁸⁵ and fully charging an empty EV battery could take anywhere from 30 minutes to 12 hours.⁸⁶

Manufacturers also targeted different segments of the EV market, reflecting diverse consumer preferences. Mercedes, BMW, and Chinese manufacturer NIO, for example, marketed luxury EVs to wealthy buyers. NIO sold a mid-size SUV that started at \$68,000, nearly twice the starting price of Tesla’s Model Y.⁸⁷ In 2017, NIO’s \$1.5 million EP9 supercar even set a speed record on Germany’s Nürburgring track, besting all previous production vehicles, from Ferrari to Lamborghini.⁸⁸ Others

⁸ The Volt was the first mass-market plug-in hybrid.

specialized in selling small, low-cost EVs, such as BYD's Seagull or Wuling's Hongguang Mini, which measured 115 inches long and started at \$4,500.⁸⁹ The Hongguang Mini sold 550,000 units in China in 2022, making it the country's best-selling EV model that year.⁹⁰ In 2023, the Chinese SUV maker Great Wall launched Ora, an EV brand targeting women that featured cars named for cats—reportedly to attract fans of Japan's *Hello Kitty* characters—and one bearing a strong resemblance to the VW Beetle.⁹¹

Government

The destructive climate impact of internal combustion engines (ICE) vehicle emissions drove governments to encourage EV adoption with a range of policy levers. Some measures disincentivized vehicle ownership. Scandinavian municipal governments limited vehicle traffic in urban centers, charging tolls for inner city entry. The Norwegian national government incentivized EV adoption beginning in the 1990s, in part to boost local Ford joint-venture EV startup Think!⁹² In 2016, Norway banned ICE vehicle sales completely as of 2025. Other countries followed suit: In 2020, the United Kingdom announced it would ban sales of new diesel and gas-powered vehicle by 2030.⁹³ In 2023, the EU voted to effectively ban the sale of ICE vehicles by 2035.⁹⁴ The Japanese government announced it would require all new cars manufactured after 2035 to be fully electric, pledging to become completely carbon neutral by 2050.⁹⁵ By 2023, Norway's greenhouse gas emissions were down 30% from 2009 levels, and EVs made up 80% of all new car sales.⁹⁶

Policymakers also set vehicle emission caps. In 2021, for example, Chinese leader Xi Jinping vowed that China would reach net zero vehicle emissions before 2060.⁹⁷ In 2023, the EU voted to reduce new vehicle emissions to 55% of 2021 levels by 2030.⁹⁸ That same year, the Biden administration proposed a set of vehicle emissions limits that would require half of the vehicles sold in the U.S. to be electric by 2030, and two-thirds to be electric by 2032.⁹⁹ The U.S. Environmental Protection Agency hoped the new standards would “accelerate the transition to a zero-emissions transportation future, protecting people and the planet.”¹⁰⁰

Governments also used supply-side subsidies to incentivize EV makers and demand-side subsidies to incentivize EV buyers, ranging from purchase grants to tax breaks and rebates. In 2023, the Korean government offered EV buyers one-time purchase subsidies of up to \$5,200.¹⁰¹ Japan offered EV purchase subsidies worth up to \$7,600 per year per customer.¹⁰² Often, EV subsidies came with strings attached. Under the Biden administration's Inflation Reduction Act, for example, U.S. buyers received a \$7,500 tax credit for purchasing EVs assembled in North America—a measure meant to reward EV purchasers and create domestic manufacturing jobs.¹⁰³ Germany's economic ministry offered car buyers a €4,500 purchase subsidy, but only on EVs that cost less than €40,000.¹⁰⁴ In 2024, Germany planned to lower purchase subsidies to €3,000.¹⁰⁵

In addition to providing subsidies and tax incentives, governments spent prolifically on research, controlling natural resources in the EV supply chain, and electrifying municipal vehicles. Goldman Sachs estimated that Chinese investment in green technology allowed Chinese firms to build EV plants three-times faster and for 80% less than equivalent plants in the U.S. that were retooled or otherwise converted from existing facilities.¹⁰⁶ From 2009 through 2022, Chinese government support for EVs reached an estimated \$173 billion (see **Exhibit 9**).¹⁰⁷ The largest share of funds went toward public procurement: between 2019 and 2021, Chinese government agencies spent more than \$15 billion on fleets of electric buses, police cars, and ambulances.¹⁰⁸ Public procurement was especially good business for BYD, which manufactured electric buses in addition to passenger vehicles.¹⁰⁹ In 2022, BYD alone received \$1.5 billion in manufacturer subsidies from the Chinese government (equal to 60% of the company's net annual profits).¹¹⁰ In 2023, subsidies briefly lapsed, and Chinese automakers braced

themselves for demand and supply chain impacts.¹¹¹ But in June, China surprised industry-watchers with a \$72 billion tax break package for manufacturers, effective from 2024 until 2027.¹¹²

Governments also played a role in building out infrastructure, most notably by developing charging stations and creating fueling networks along national transportation routes, a key piece of infrastructure required for EV adoption (see **Exhibit 10**). Consumer anxiety about running out of battery power was a leading reason for reluctance to consider purchasing an EV; areas that had more chargers^h saw a corresponding increase in EV purchases.¹¹³ However, not all chargers worked for different EVs, and without standardization, some areas saw patchworks of EV chargers that failed frequently, required different payment methods, or were exclusive to certain brands; governments sometimes intervened to set industry standards.¹¹⁴ (**Appendix B** details EV charging infrastructure.)

In some instances, strategic competition between governments influenced EV adoption. In the aftermath of Russia's invasion of Ukraine, for example, NATO allies levied sanctions which forbade American and European brands from selling EVs to Russian drivers. Chinese EV makers rushed to fill the vacuum.¹¹⁵ Meanwhile, Hyundai's sales in China fell by more than 50% between 2017 and 2021, after China restricted Korean imports to retaliate against the Korean government.^{116,i}

Consumers

In 2022, spending on EVs worldwide reached \$425 billion, up 50% from 2021. Governments accounted for 10% of spending; consumers accounted for the rest.¹¹⁷ EV buyers wanted assurance that their vehicles had sufficient range to reach their destinations. More granular consumer preferences (on-deck technology, size, price) varied by country market.

In China, analysts identified user experience, convenience, and price as EV buyers' top priorities.¹¹⁸ Whether a car was electric or gas-powered was secondary.¹¹⁹ Analysts highlighted the importance of sophisticated on-board technology: "[. . .] Chinese consumers are more keen on new [in-car] technologies that can bring [them] comfort and entertainment [. . .]."¹²⁰ One study found that each of the top 10 most desirable EVs in China had voice command functionality, and that top-selling Chinese EVs understood at least 87% of Chinese drivers' voice commands.¹²¹ Tesla's Model Y voice command feature, by contrast, underperformed for drivers with certain regional accents.¹²² "[Teslas] are not designed for Chinese drivers and passengers, and they are not the up-to-date smart vehicles that many rich Chinese consumers desire," an analyst noted.¹²³ Technology was an important differentiator in the Chinese market—a few Chinese EVs boasted large screens and karaoke systems.¹²⁴ One Chinese car buyer remarked, "Chinese-made smart electric cars are now good choices for middle-class consumers because their technologies and services cater to our demands."¹²⁵

European consumers prioritized size and name recognition. BYD's head of marketing in Britain said that the company designed its Atto 3 SUV style to maximize space and elevate the driver's position inside the vehicle.¹²⁶ "People in Germany," however, "buy cars based on the brand," a German auto industry expert pointed out. "The brand is decisive."¹²⁷

Korean drivers cited charging issues as the primary barrier they faced to EV adoption.¹²⁸ In a 2021 survey, 60% of respondents were concerned about charging infrastructure and 30% were concerned about charging times.¹²⁹ Nevertheless, most Koreans were optimistic about the EV industry: 85%

^h A charger provided power to one EV; a charging station contained one or more chargers.

ⁱ The move came after Korea announced it would deploy a U.S. anti-missile system along its borders, aligning it with China's strategic rivals, the U.S. and Japan.

expected EVs to grow in popularity, and 75% believed that EV adoption was a crucial environmental protection measure.¹³⁰

In the U.S., EVs polarized car buyers. In 2022, EVs made up 5.8% of new vehicle sales in the U.S. By the first quarter of 2023, that number rose to 7.8%.¹³¹ Despite their growing popularity, half of Americans surveyed said that they would not consider an EV the next time they purchased a car, 38% responded that they would, and 13% responded that they did not plan on purchasing a vehicle.¹³² In the U.S., age, location, and political party skewed attitudes toward EV adoption.¹³³ Younger people, Democrats, and city-dwellers were more likely to purchase an EV.¹³⁴ (See **Exhibit 11**.) Over two-thirds of this group was very likely or somewhat likely to consider an EV for their next vehicle purchase.¹³⁵ Net enthusiasm for EVs was growing: between 2022 and July 2023, the percentage of Americans who worried that EVs were too expensive fell from 49% to 42%, the percentage worried about battery range fell from 44% to 39%, and the percentage worried about finding a place to charge fell from 35% to 33%.¹³⁶ In 2023, 72% of Americans interested in purchasing an EV cited their environmental benefits and 70% cited the price of gas.¹³⁷

Markets and Manufacturers Outside of China

In Europe

In 2022, the European EV market grew 15% year-over-year; in contrast, the Chinese EV market grew 82%.¹³⁸ European EV sales totaled 2.6 million units; Chinese EV sales totaled 6.1 million.¹³⁹ German manufacturers led the fast-growing global market for luxury (or “premium”) EVs, which in 2022 grew 95% year over year, compared to 57% growth in the mass-market segment.¹⁴⁰ “Europe is a premium export market,” Ford’s Farley remarked, “[Chinese automakers are] all going there.”¹⁴¹

The market also attracted manufacturers of mass-market vehicles. In 2022, BYD launched three new vehicles in Germany and reached an agreement with Germany’s largest car rental company to sell it 100,000 BYD vehicles by 2029.¹⁴² As China made inroads, one report estimated that, by 2030, Chinese EVs could cost European brands \$7.7 billion annually in lost profits.¹⁴³ China was not Europe’s only worry: in 2022, Tesla opened a production facility outside Berlin, its first in Europe.¹⁴⁴ By early 2023, Tesla accounted for one in every five new cars registered in Germany.¹⁴⁵ Other U.S. automakers fought for market share in Europe, albeit less successfully. GM’s German subsidiary, Opel, operated at a loss for 10 years before GM sold the brand in 2017.¹⁴⁶ In 2022, Ford’s sales in Germany came to around 25% of VW’s.¹⁴⁷ Ford began shrinking its footprint in Europe—in 2023, the carmaker entered talks to sell a factory in southwest Germany.¹⁴⁸ BYD was among the companies interested in buying the plant.¹⁴⁹

Volkswagen, headquartered in Germany, was the world’s largest automaker by revenue in 2022, with almost \$294 billion in revenue from the sale of 4.6 million vehicles.¹⁵⁰ That year, VW was also Europe’s largest carmaker and its best-selling EV brand, with a 20% market share via the sale of 358,000 EVs.¹⁵¹ The company pledged to halt production of non-electric vehicles in Europe by 2033, two years before the EU moratorium on gas-powered vehicle sales took effect.¹⁵²

VW ranked fourth in global EV sales,¹⁵³ but its status as leader masked declining market share: between 2021 and 2023, its share of Europe’s EV market shrank about 17%.¹⁵⁴ Worldwide, VW’s share of the passenger EV market fell from 14% in the second quarter of 2021 to 7% in the first quarter of 2023.¹⁵⁵ Tesla’s global market share hovered around 15% throughout the period, while BYD’s tripled to 21% (refer to **Exhibit 2**).

Analysts attributed VW's declining market share in part to its software-development unit, Cariad.¹⁵⁶ The Artemis, to be sold under VW's premium brand, Audi, was pitched as VW's answer to Tesla.¹⁵⁷ In 2022, however, Cariad delayed the Artemis launch by three years, to 2027.¹⁵⁸ In 2023, rumors circulated that mismanagement at Cariad was likely to delay the launch even further. Software issues also plagued VW's other EVs, ranging from sudden braking to faulty displays.¹⁵⁹

In 2023, as part of a bid to improve its EV technology and boost sales in China, VW paid \$700 million for 5% ownership in XPeng.¹⁶⁰ In 2026, VW and XPeng planned to launch two brand-new EV models in China under the VW brand.¹⁶¹ Meanwhile, Audi also partnered with the Chinese carmaker SAIC to develop a line of EVs for the Chinese consumer market.¹⁶²

BMW sold more units than any other luxury EV maker, and controlled 21% of the luxury EV market worldwide.¹⁶³ It ranked 13th in the overall global EV market.¹⁶⁴ In 2022, BMW announced that by 2025, it aimed to have sold a total of two million battery EVs, and by 2030 to have half of its global sales come from battery EVs.¹⁶⁵ Several months later, BMW's CEO told investors BMW would hit its EV sales targets "well ahead of schedule."¹⁶⁶ He also expected BMW to bring a pathbreaking hydrogen-powered vehicle to market in the next two to seven years.¹⁶⁷ Still, BMW never promised to eliminate gas-powered vehicles from its lineup entirely. As part of its "flexible powertrain" strategy, the company strategically sold battery electric, gas-powered, and plug-in hybrid versions of its popular models, allowing customers to choose which version they preferred.¹⁶⁸

Analysts forecast disruption in the worldwide luxury EV market, warning that BMW's top spot was far from assured.¹⁶⁹ In addition to competition from Mercedes-Benz (see below), Audi, and Volvo, BMW also faced competition from emerging players. The market cap of NIO, the Chinese luxury EV brand founded in 2014, briefly surpassed BMW's in 2020,¹⁷⁰ and in 2022, NIO opened its first European showroom in Berlin, putting German automakers on notice.¹⁷¹ That year, NIO announced plans to bring its innovative battery-swapping network to Europe, allowing drivers to save time otherwise spent recharging their EVs by simply exchanging depleted batteries for charged batteries at swap stations.¹⁷² Then, in June 2023, NIO beat BMW to the market with its ET5 Touring electric station wagon.¹⁷³ BMW had previously billed its i5 Touring, set to launch in 2024, as Europe's first EV "premium" wagon.¹⁷⁴ "We are comprehensively ahead of our competitors in terms of products and services," NIO's co-founder declared.¹⁷⁵ Within China, NIO controlled 33% of the luxury EV market, compared with BMW's 23%. Another Chinese firm, Zeekr, held 20%.¹⁷⁶

BMW's early forays into EVs included its miniature i3, launched in the U.S. in 2014. Despite its novel looks and lively performance, at their peak, U.S. sales of the i3 between 2014 and 2022 barely cracked 11,000 per year.¹⁷⁷ The company discontinued U.S. sales that year after producing a total of 250,000 units worldwide.¹⁷⁸ BMW had loftier expectations for a new, cost-cutting electric vehicle platform,^k called Neue Klasse.¹⁷⁹ New car platforms could take as long as seven years to design and build,¹⁸⁰ and in 2010 one analyst pegged the build costs alone at about \$1 billion.¹⁸¹ BMW executives hoped that the company's new EV platform could reduce production costs by 30%.¹⁸² In 2022, BMW's CEO announced that the company's Neue Klasse vehicles would become "as profitable as vehicles with state-of-the-art combustion engines." The CEO added that batteries for BMW's new line of EVs would include "new

^j Although the first hydrogen engines appeared in 1806, no automaker had yet seen mass market success with a hydrogen vehicle, partly due to the lack of fueling infrastructure. See Fastech, "The Past, Present, and Future of Hydrogen Vehicles: 2023 Update," April 4, 2023, <https://tinyurl.com/3zv6tjw>, accessed August 2023.

^k Vehicle platforms were the basic structures that manufacturers fitted components to – like a car's skeleton. It was more efficient to assemble EVs using dedicated EV platforms, which carmakers optimized for EV components, than repurposed platforms from gas-powered vehicle models.

cell chemistry and new cell formats,” reportedly including material boosts to energy density, charging speed, and range, while lowering cost by 50%.¹⁸³ In 2022, BMW produced its own batteries in four factories worldwide and was investing \$1.7 billion in two new battery plants in the U.S.¹⁸⁴

Mercedes-Benz was the world’s second-best selling luxury EV brand.¹⁸⁵ The company sold 2.4 million vehicles annually (including ICE vehicles) and controlled 16% of the premium EV market worldwide.¹⁸⁶ Mercedes was a venerated brand; Carl Benz, the company’s founder, patented the world’s first gas-powered automobile in 1886.

In 2022, Mercedes CEO Ola Källenius announced he would invest \$45 billion to transform Mercedes into the world’s top luxury EV brand and ensure that the brand’s vehicles were 100% electric before 2030: “We’ve always been a leader in innovation and technology and in timeless luxury, luxury in its time. What will be Mercedes’ quest now going into the electrical era and the digital era is to deliver again on those two.”¹⁸⁷ He told reporters, “[T]he competitive intensity is increasing. It’s the most fun time to work in automotive since 1886. It’s also the most uncertain time.”¹⁸⁸

Källenius said, “[W]e have to realize that the transition to zero-emission driving is really a marathon [. . .] it’s not something we should look at as a sprint.”¹⁸⁹ Mercedes struggled with the quality of its EV batteries, supplied by the Chinese company Farasis.¹⁹⁰ In 2021, a Mercedes executive described the situation as a “catastrophic” snag for its global strategy.¹⁹¹ In 2023, making matters worse, Mercedes misjudged the EV market in China and had to slash the price of its flagship EV by close to \$33,000.

In addition, EV analysts expressed concerned that VW, Mercedes, and BMW lacked the software chops to compete with their younger, savvier rivals. ICE vehicles required less advanced software to manufacture than EVs, which relied on software to optimize charging, monitoring, and even braking systems. An industry expert explained, “From a hardware perspective I would have no doubts that they can make superb cars. But can the complexity and quality needed for the software be mastered by a 120-year-old hardware manufacture? I’m not so sure.”¹⁹² In 2022, Mercedes ranked 16th in their share of the global EV market, with 1.9%.¹⁹³ They were a leader in autonomous driving features, becoming the first carmaker to receive U.S. and European government approval for “Level 3” driving features, in which drivers were able to turn over more control than featured in rival Tesla’s Autopilot.¹⁹⁴

East Asia: Japan and South Korea

Toyota, the largest carmaker in the world by units sold, was based in Japan.¹⁹⁵ The company sold 10.5 million vehicles in 2022, generating \$267 billion in revenue, and aimed to boost sales by 15% in 2023.¹⁹⁶ From 2009 until 2023, Toyota was helmed by Akio Toyoda, who viewed EVs as “overhyped,” arguing that their cost and inadequate charging infrastructure made mass EV adoption unlikely.¹⁹⁷

Rather than actively compete with a portfolio of BEVs and PHEVs, Toyota had pioneered hybrid electric vehicles (HEVs). In the late 2010s, as Toyota’s competitors rushed to electrify their product offerings, Toyota positioned itself as the last major auto company “straddling combustion and electric propulsion.”¹⁹⁸ It led the global HEV market in 2022 with 5 million Priuses sold since 1997.¹⁹⁹

In January 2023, Koji Sato became CEO. He told staff that Toyota needed to “drastically change the way we do business,” and assured buyers that Toyota would adopt an “EV-first mindset” going forward.²⁰⁰ That meant investing in a battery parts supply chain and assembling its own EV batteries.²⁰¹

In July 2023, Toyota took investors by surprise, announcing that it had already achieved a breakthrough in EV battery technology: its researchers developed a battery that took 10 minutes to

charge and boasted a 745-mile range.²⁰² “Toyota just became a force to reckon with [. . .]. For the first time in the history of mass-production EVs, a battery-powered car will have the same driving range as one with an engine and a gas tank,” one industry publication trumpeted.²⁰³ Analysts expected Toyota’s new batteries to hit the market in 2027.²⁰⁴

Toyota was making up for lost time. “We need to increase our speed and efforts to firmly meet the customer expectations in the Chinese market,” Sato stressed, after an appearance at the 2023 Shanghai Auto Show.²⁰⁵ In China that year, Toyota reduced the price of some models by 15% and launched its first fully electric sedan, the bZ3.²⁰⁶ The bZ3 ran on battery technology that Toyota bought from BYD, and pricing for the sedan began at \$27,000.²⁰⁷ However, three months after launch, Toyota recalled 12,000 bZ3 units, citing an issue with the model’s rear doors.²⁰⁸

Hyundai, South Korea’s largest carmaker, manufactured South Korea’s best-selling EV, the Ioniq 5.²⁰⁹ In 2022, the Ioniq sold 27,399 units in Korea (Tesla’s Model Y sold 6,073)²¹⁰ and 510,000 Ioniq units worldwide, marking a 41% increase over the company’s 2021 sales and making Hyundai the sixth best-selling global EV maker.²¹¹ In May 2023, Hyundai’s CEO announced an ambitious new strategy: the company would grow its productive capacities, invest over \$18 billion in the South Korean EV market, and become one of the world’s top three best-selling EV makers by 2030.²¹²

The success of Hyundai’s global strategy, like Toyota’s, depended on sales to the Chinese EV market. Hyundai’s CEO said, “We have a joint venture [with Beijing carmaker BAIC] in China. We are now on a deep dive on how we can regain the competitiveness of the China market.”²¹³ He added, “I think the first step that we’re looking at is how we can optimize the operational capacity in China. And the next step should be our focus on the product portfolio, which should be attractive to local customers with the comparable software functions, as well as hardware and design features.”²¹⁴ Hyundai’s joint venture sold 123,259 EVs in China in the first half of 2023, a 13% increase over sales the year before.²¹⁵

Hyundai also invested in joint ventures in the U.S. In 2022, Hyundai invested \$5.5 billion in a battery plant and partnered with the Korean battery-maker SK to assemble EV batteries in the U.S. state of Georgia.²¹⁶ The agreement was backed by \$1.8 billion in tax breaks and other incentives.²¹⁷ SK already operated one battery production facility in a suburb of Atlanta, Georgia. There, SK manufactured batteries for Ford and VW.²¹⁸

United States

Tesla was founded by engineers Martin Eberhard and Marc Tarpenning in California in 2003.²¹⁹ Elon Musk became Tesla’s chairman in 2004,²²⁰ its CEO in 2008, and oversaw its initial public offering in 2010.²²¹ A decade later, the press called Tesla the “king of EVs” and “the world’s favorite EV brand.”²²² It was worth \$76 billion, and its Model Y was the world’s best-selling EV.²²³ The latter sold 771,000 units in 2022 (the second best-selling EV model was BYD’s Song Plus, which sold 477,000 units).²²⁴ In 2022, Tesla delivered 1.3 million vehicles worldwide, a 40% increase over 2021 deliveries.²²⁵

Then, between April 2022 and January 2023, Tesla lost 65% of its market value.²²⁶ Analysts cautioned Tesla’s investors that the company faced serious challenges, not least among them its CEO’s fraught Twitter acquisition and controversial public appearances.²²⁷ In addition, one researcher explained, “Tesla has continued to try and grow at very aggressive rates. 30 percent plus this year, 30 percent plus next year. And it has no new product offerings. How do you create incremental demand, when you don’t have anything new per se? That’s Tesla’s challenge over the next four to six quarters.”²²⁸ By the end of 2026, one forecast had Tesla’s global market share dropping to 18% and the shares of the market controlled by Ford and GM would increase to 14% each on the back of new product

launches.²²⁹ In May 2023, Musk announced that he would continue to prioritize sales volume at the expense of near-term profit, making some investors uneasy.²³⁰

Tesla investors still had reason to be hopeful. In 2023, Tesla remained the leading EV maker in the U.S., and the most valuable EV maker in the world. In the second quarter of 2023, U.S. drivers bought a record 295,000 EVs, a 48% year-over-year increase.²³¹ Tesla's Model Y accounted for more than 105,000 sales, whereas just six EV models from legacy U.S. automakers sold more than 5,000 units each.²³² In 2023, the company was preparing to launch its Cybertruck, a direct competitor to Ford's line of EV pick-ups.²³³ Analysts also found that Tesla's American competitors were overstocked—excluding Tesla, U.S. automakers had the equivalent of 70 sales-days of EV inventory on their lots in 2023.²³⁴ Overstock created downward pressure on prices.²³⁵ Tesla, by contrast, built vehicles to order.²³⁶ *Barron's* warned, “[N]on-Tesla EV inventories aren’t an issue yet, but they could become one.”²³⁷

Tesla also boasted a network of more than 45,000 proprietary “Superchargers,” “the largest global, fast charging network in the world.”²³⁸ The company claimed that 15 minutes at a Supercharger station would give Tesla vehicles 200 miles of range.²³⁹ From 2012, the year Tesla introduced the Supercharger, until 2023, only Teslas could use Superchargers, which comprised 60% of public fast chargers in the country.²⁴⁰ In the U.S., its competitors used the more common Combined Charging System.²⁴¹ (Refer to **Appendix B**.) Then, in June 2023, Ford, GM, and Volvo all announced they would migrate to the Supercharger network, making Tesla the presumptive industry standard.²⁴² Tesla did not demand payment from competitor companies but benefited from drivers paying for electricity and, some speculated, from the data it could glean from a larger pool of cars plugging into its ports.²⁴³

Ford announced in 2023 that it was “deep into development of [its] second-generation EVs,” including a successor to its popular F-150 Lightning electric truck.²⁴⁴ Ford specialized in EV trucks, which it manufactured at a purpose-built, carbon-neutral facility in the U.S. state of Tennessee.²⁴⁵ Ford produced 150,000 Lightning units annually, and planned to increase production of its electric Mustang Mach-E to 200,000 units per year.²⁴⁶ The Lightning included an electric generator for off-road power that could briefly power homes during electricity outages.²⁴⁷

To scale up production and compete with Tesla on price, in CEO Farley's words, Ford “radically simplified” its EVs.²⁴⁸ Ford would minimize the number of body styles it offered and the number of components in its vehicle platforms.²⁴⁹ “Imagine three body styles, each with volume potential of up to one million units, and just a handful of orderable combinations,” Farley told reporters.²⁵⁰ Ford also shrank the footprint of its EV batteries, he added, “for competitive size.”²⁵¹ Farley also said that Ford, unlike Tesla, was not willing to pursue sales growth “at any cost.”²⁵² Nevertheless, he estimated that Ford's 2023 losses on EVs would total \$3 billion, comparing Ford's EV unit to a startup subsidized by Ford's gas-powered vehicle sales.²⁵³

A Chinese manufacturer, CATL, produced batteries for Ford.²⁵⁴ CATL also sold batteries to Ford's competitors, including BMW and GM.²⁵⁵ In 2023, Ford entered a licensing agreement with CATL to produce the batteries itself, with help from Chinese engineers, at a \$3.5 billion battery plant in Michigan. Some U.S. politicians criticized Ford for going into business with the Chinese battery-maker, potentially giving Chinese business interests the chance to influence U.S. energy and automotive policy.²⁵⁶ Ford published a statement defending its arrangement with CATL, writing that on the contrary, its factory would help the U.S. “compete and win globally.”²⁵⁷

GM sold 20,670 EVs in the first quarter of 2023²⁵⁸ and 15,000 EVs in the second quarter of 2023.²⁵⁹ Combined, GM's EV sales grew 365% year-over-year in the U.S. in the first six months of 2023, more

than any other EV maker.²⁶⁰ “GM is off to a great start in 2023,” the company’s North America president announced in April. That year, its new product launches included the Chevrolet Blazer and the Equinox, both assembled at a facility in Mexico.²⁶¹ GM was on track to launch several additional EVs in 2024, which the company’s leadership hoped would grow GM’s sales volume.²⁶² These included the 2024 Silverado, an electric pick-up to compete with Ford’s F-150 Lightning, and the much-hyped Escalade EV, sold under GM’s luxury Cadillac brand and manufactured at a plant in Detroit, on which the company spent \$2.2 billion optimizing for EV production.²⁶³ Cadillac was also preparing to release a customizable EV, the Celestiq, which would retail for over \$300,000.²⁶⁴ Barra announced that the automaker was set to produce 400,000 EVs by mid-2024, and 1 million EVs by 2025.²⁶⁵

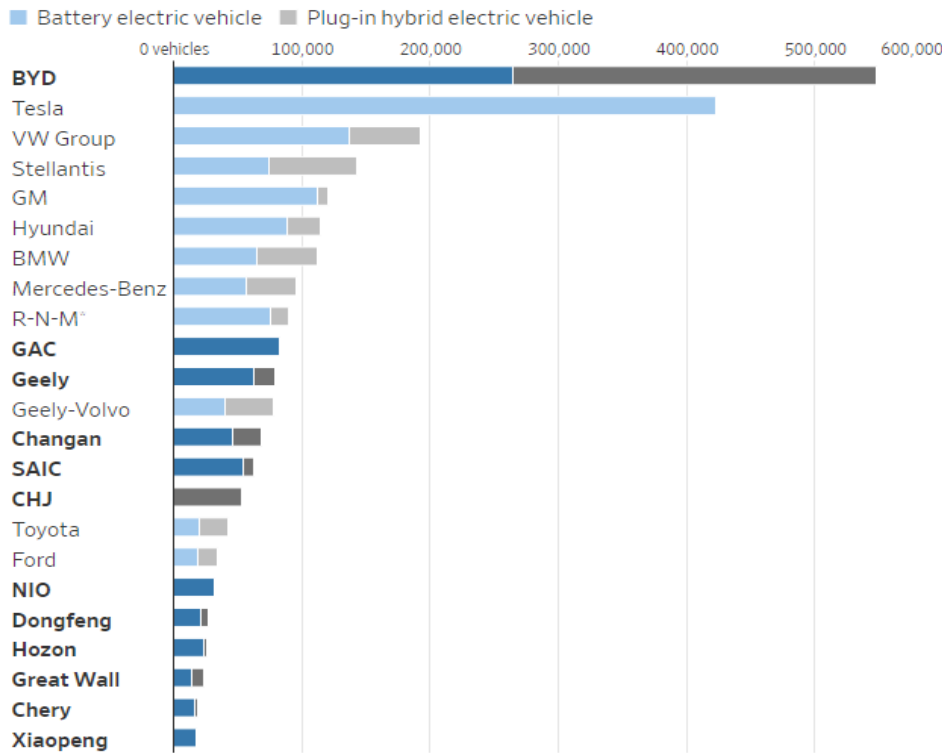
Barra neglected to mention that one line, GM’s flagship EV, the Chevy Bolt, accounted for 19,700 of GM’s 20,670 Q1 2023 EV sales.²⁶⁶ The Bolt, introduced in 2016, sold 15,000 units in 2021 and more than 16,000 units in 2022.²⁶⁷ GM pitched the Bolt as “America’s most affordable EV” – 2023 models began at approximately \$28,000.²⁶⁸ Yet the Bolt had faced challenges. A battery fire in 2021 led to a recall of Bolt EVs in the U.S. and Canada, a temporary halt to production, \$6,000 refunds to previous Chevy owners who agreed not to sue over the battery recall, and a subsequent lowering of the model’s 2023 price.²⁶⁹ In April 2023, Barra announced GM would discontinue the Bolt before the end of 2023.²⁷⁰ Analysts, skeptical of GM’s plans to scrap its most popular EV, also criticized the company for not scaling production of other, newer EV models.²⁷¹ In July, 2023, following strong Bolt sales, Barra announced that the company had reversed plans to retire the Bolt and that Bolts henceforth would be built on the company’s new Ultium battery platform “that’s really going to help us get costs down.”²⁷²

GM began producing its “Ultium” EV batteries in 2022 at plants in the U.S. states of Michigan, Ohio, and Tennessee in partnership with the South Korean electronics company LG.²⁷³ GM’s Ultium battery was designed to minimize capital investment and production costs.²⁷⁴ By 2025, Barra promised, GM’s battery technology would enable the company to sell more EVs than Tesla.²⁷⁵ But Ultium batteries were difficult to produce – in the first two quarters of 2023, GM sold approximately 2,800 EVs equipped with Ultium batteries, a fraction of the EVs that GM produced during that period, falling short of expectations.²⁷⁶ In June, Barra wrote GM shareholders to assure them that the company was “taking steps to lower our capital spending,” and that GM would lower operating costs by an additional \$1 billion before the end of 2024.

Conclusion

In 2023, racing to control a brand-new market, EV makers faced challenges reminiscent of those faced by Ford’s Henry Ford and GM’s Alfred Sloan a century earlier. At the same time, the EV industry was innovating faster than any other sector of auto market; in 2021, the number of new gas-powered vehicle patents fell 24%, whereas the number of new EV patents rose 44%.²⁷⁷ Berkshire Hathaway’s vice chairman said of BYD’s Wang, “This guy is a combination of Thomas Edison and Jack Welch—something like Edison in solving technical problems, and something like Welch in getting done what he needs to do. I have never seen anything like it.”²⁷⁸

Could the “old legends” match these innovative EV startups? How credible were the responses of large incumbent players to Tesla and the sudden prominence of Chinese EV makers? The stakes had only increased as Chinese firms began marketing their EVs in Europe and elsewhere in the world. In a generation where climate change had spurred pledges to decarbonize and phase out gasoline-powered vehicles, how would the opportunity of a lifetime play out for firms in the automotive industry?

Exhibit 1 Q1 2023 Global EV Sales, Combined Battery and Plug-In Hybrid EVs (units sold)

Source: EV Volumes data in Greg Ip, "China's EV Juggernaut Is a Warning to the West," *The Wall Street Journal*, June 7, 2023, <https://www.wsj.com/articles/chinas-ev-juggernaut-is-a-warning-for-the-west-1389f718>, accessed July 2023.

Note: Chinese EV makers are listed in bold. R-N-M: Renault-Nissan-Mitsubishi.

Exhibit 2 Global EV Market Share of Market Leaders: BYD, Tesla, Volkswagen (Q2 2021-Q1 2023)

Brands	Q2 2021	Q3 2021	Q4 2021	Q1 2022	Q2 2022	Q3 2022	Q4 2022	Q1 2023
BYD Auto	7%	11%	12%	14%	16%	20%	20%	21%
Tesla	15%	15%	14%	15%	12%	13%	12%	16%
Volkswagen	14%	11%	10%	8%	8%	8%	8%	7%

Source: Counterpoint Research, "Global Passenger Electric Vehicle Market Share," June 13, 2023, <https://www.counterpointresearch.com/global-electric-vehicle-market-share/>, accessed August 2023.

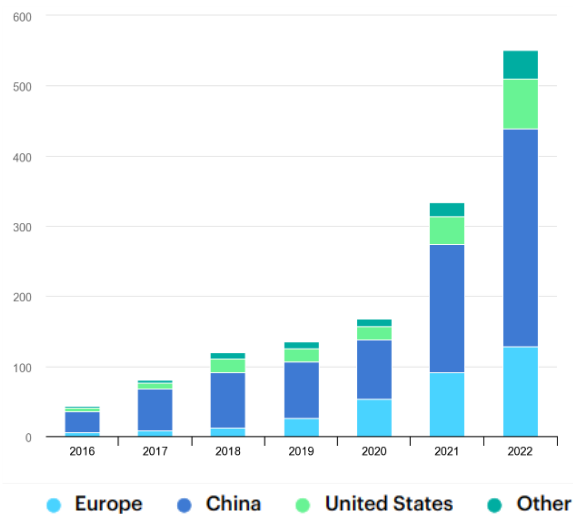
Note: EVs include BEVs and PHEVs and not HEVs or fuel cell vehicles (refer to **Appendix A**).

Exhibit 3 Automaker Production & Revenue Data, by Market Capitalization, Year-End 2022

Company (Headquarters)	Market Cap (\$bn)	EVs Sold ('000)	All Sold Vehicles ('000)	EV Share (%)	EV Revenue (\$bn)	Total Vehicle Revenue (\$bn)	Total Company Revenue (\$bn)
Tesla (U.S.)	\$389	1,314	1,314	100%	\$78	\$78	\$82
Toyota (Japan)	\$187	2,730	9,567	29%	\$19	\$236	\$270
BYD (China)	\$95	112	1,890	6%	\$62	\$329	\$424
Volkswagen	\$72	572	8,200	7%		\$225	\$298
Mercedes-Benz	\$70	319	2,044	16%		\$138	\$160
BMW (Germany)	\$58	434	2,400	25%		\$132	\$152
General Motors	\$48	39	5,900	<1%		\$144	\$157
Ford (U.S.)	\$47	62	4,231	2%		\$149	\$158
Stellantis	\$46	288	5,840	5%		\$190	\$192
Honda (Japan)	\$39	0	4,080	0%		\$81	\$124
Maruti Suzuki	\$31	0	1,645	0%		\$31	\$14
Great Wall (China)	\$30	610	1,068	57%		\$20	\$20
Hyundai (South)	\$25	175	3,945	4%		\$144	\$113
Rivian (U.S.)	\$17	20	20	100%	\$2	\$2	\$2
NIO (China)	\$16	123	123	100%		\$7	\$7
Geely (China)	\$15	329	1,433	23%		\$22	\$22
Guangdong Auto	\$14	310	2,480	13%		\$16	\$16

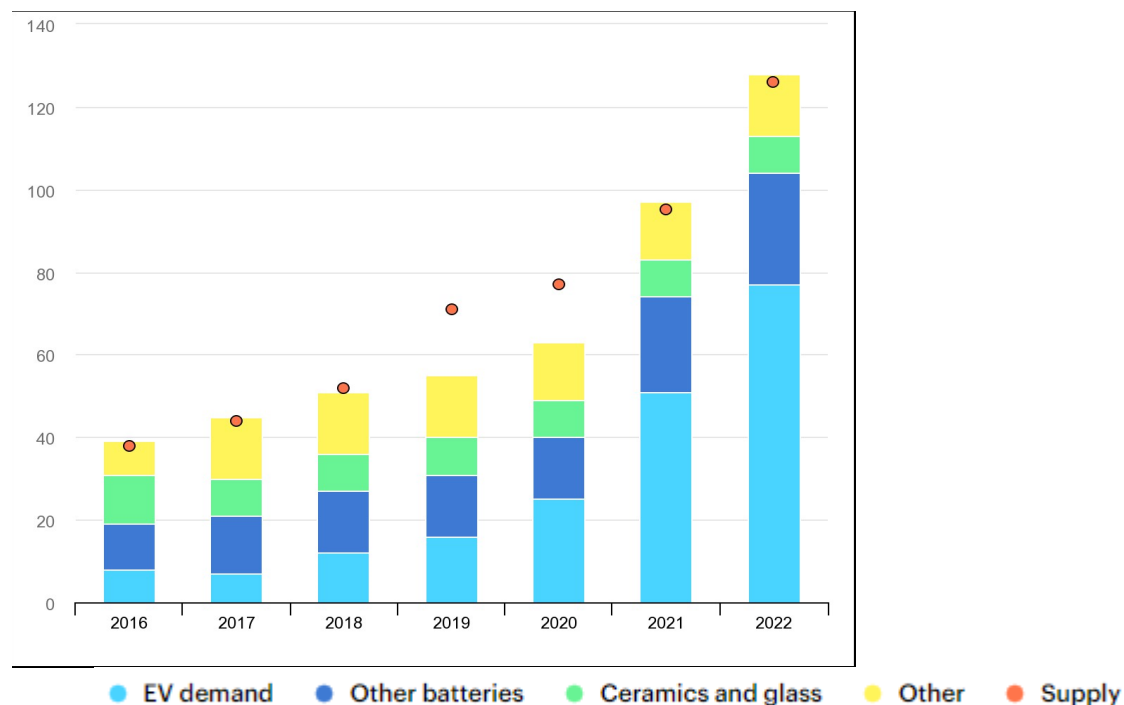
Source: "Tesla – CIQ Financial Data – Segment Data," S&P CapitalIQ, Inc., a division of S&P Global, Inc.; "Toyota Unveils New Technology that Will Change the Future of Cars," Toyota, June 13, 2023, <https://shorturl.at/ejwG8>; Mark Kane, "BYD Closes Year 2022 with Record Plug-In Car Sales in December," InsideEVs, January 4, 2023, <https://shorturl.at/adhW8>; Mark Kane, "Volkswagen Group Global BEV Sales in 2022 Reached 572,100," InsideEVs, January 18, 2023, <https://shorturl.at/exFHM>; Mark Kane, "Mercedes-Benz More Than Doubled All-Electric Car Sales in Q4 and 2022," InsideEVs, January 19, 2023 <https://shorturl.at/gpIY5>; "BMW Group Stays on Track for Success in 2022: Maintains No. 1 Position in Global Premium Segment, Doubles Electric Sales," BMW Group, January 10, 2023, <https://shorturl.at/mwORX>; Mark Kane, "US: Ford Tripled BEV Sales in December, Doubled in 2022," InsideEVs, January 7, 2023, <https://shorturl.at/vIY38>; "Stellantis Delivers Record Full Year 2022 Results; Global BEV Sales Up 41%," Stellantis, February 22, 2023, <https://shorturl.at/ejHN4>; "Honda Announces Next Steps in Preparation for U.S. EV Production," Honda, March 14, 2023, <https://shorturl.at/jnyZ4>; BS Web Team, "Maruti Takes an Offensive Stance in EV Space, to Launch 6 EVs by 2030-31," Business Standard, July 6, 2023, <https://shorturl.at/kGJ58>; "Focusing on New Energy," GWM Achieves Sales of Over One Million in 2022," GWM News, January 13, 2023, <https://shorturl.at/uOQRY>; Hyundai, "Hyundai Motor Reports 2022 Global Sales and 2023 Goals," January 3, 2023, <https://shorturl.at/ceBGT>; Mark Kane, "Rivian Set Record EV Production and Deliveries in Q4 2022," InsideEVs, January 4, 2023, <https://shorturl.at/oBLPZ>; NIO, "NIO Inc. Provides December, Fourth Quarter and Full Year 2022 Delivery Update," January 1, 2023, <https://shorturl.at/biFKY>; Geely, "Geely's Total Sales Reached 1.43 Million Units in 2022," January 10, 2023, <https://shorturl.at/fqOZ3>; Threza Gabriel, "GAC Group: 309,548 New Energy Vehicles to Be Sold in 2022, Up 116.7% Year-on-Year," TechGoing, May 1, 2023, <https://shorturl.at/pUY13>; all accessed August 2023.

Note: EVs refer BEV and PHEVs and not HEVs. Empty fields indicate unavailable data.

Exhibit 4a Automotive Lithium-Ion Battery Demand, GWh/year, by region, 2016 to 2022

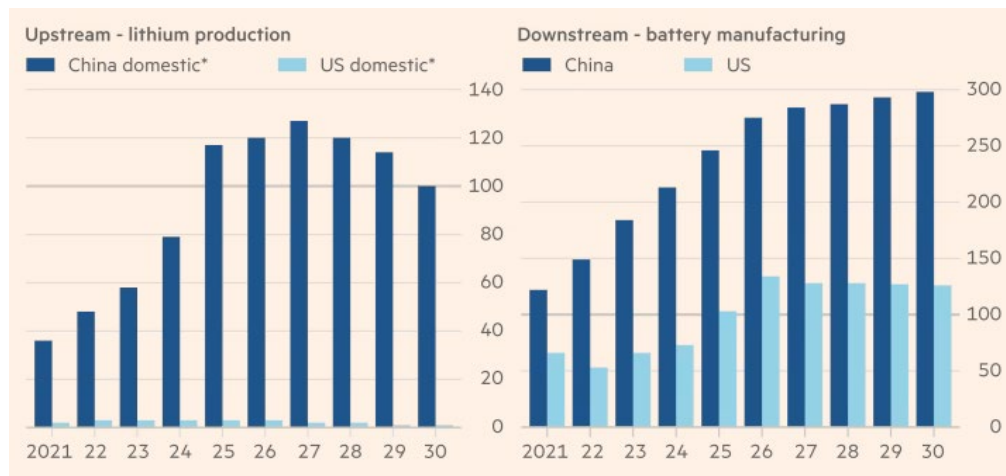
	Europe	China	U.S.	Other	Total
2016	4.8	30.3	6.2	2.5	43.8
2017	7.8	61.0	7.7	4.2	80.7
2018	12.0	79.5	18.9	9.0	119.4
2019	25.3	80.5	19.0	9.8	134.6
2020	53.0	84.9	18.6	11.2	167.7
2021	91.7	182.1	39.3	20.9	334.0
2022	127.7	312.0	70.6	40.3	550.6

Source: Adapted from IEA, Battery Demand by Region, 2016-2022, (Paris: International Energy Agency, 2023), License: CC BY 4.0, <https://tinyurl.com/2y5enzuw>, accessed August 2023.

Exhibit 4b Overall Supply and Demand of Lithium for Batteries, by sector, 2016 to 2022

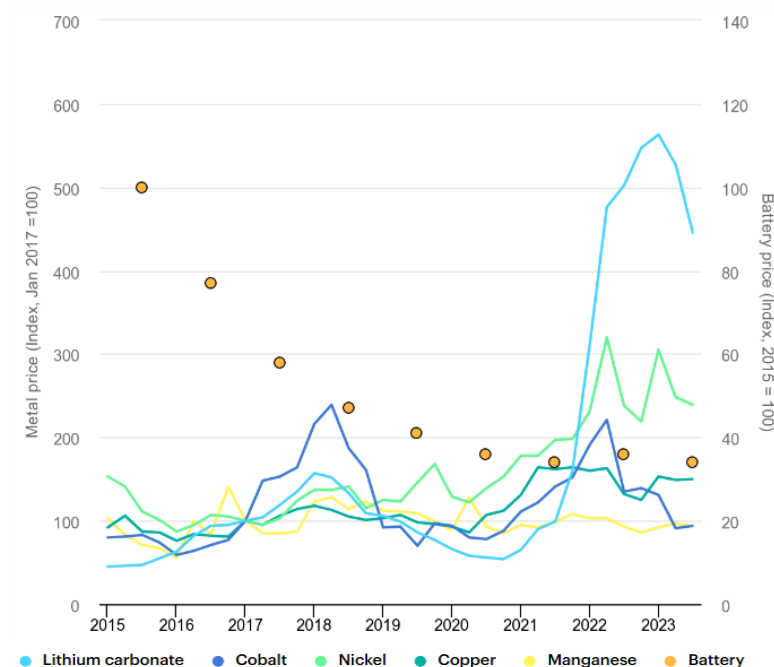
Source: IEA, Overall supply and demand of lithium for batteries by sector, 2016-2022 (Paris: International Energy Agency, 2023), License: CC BY 4.0, <https://tinyurl.com/mr3j3ebm>, accessed August 2023.

Note: Dots indicate supply.

Exhibit 4c Estimated China and U.S. Lithium Production and Battery Manufacturing, 2021-2030

Source: Edward White, "How China Cornered the Market for Clean Tech," *Financial Times*, August 9, 2023, <https://on.ft.com/3YEI9BJ>, accessed August 2023.

Note: Supply/demand estimated (%; 100 = self-sufficiency). Does not include global equity stakes or free-trade agreements.

Exhibit 4d Indexed Prices of Selected Battery Materials and Lithium-Ion Batteries, 2015-2023

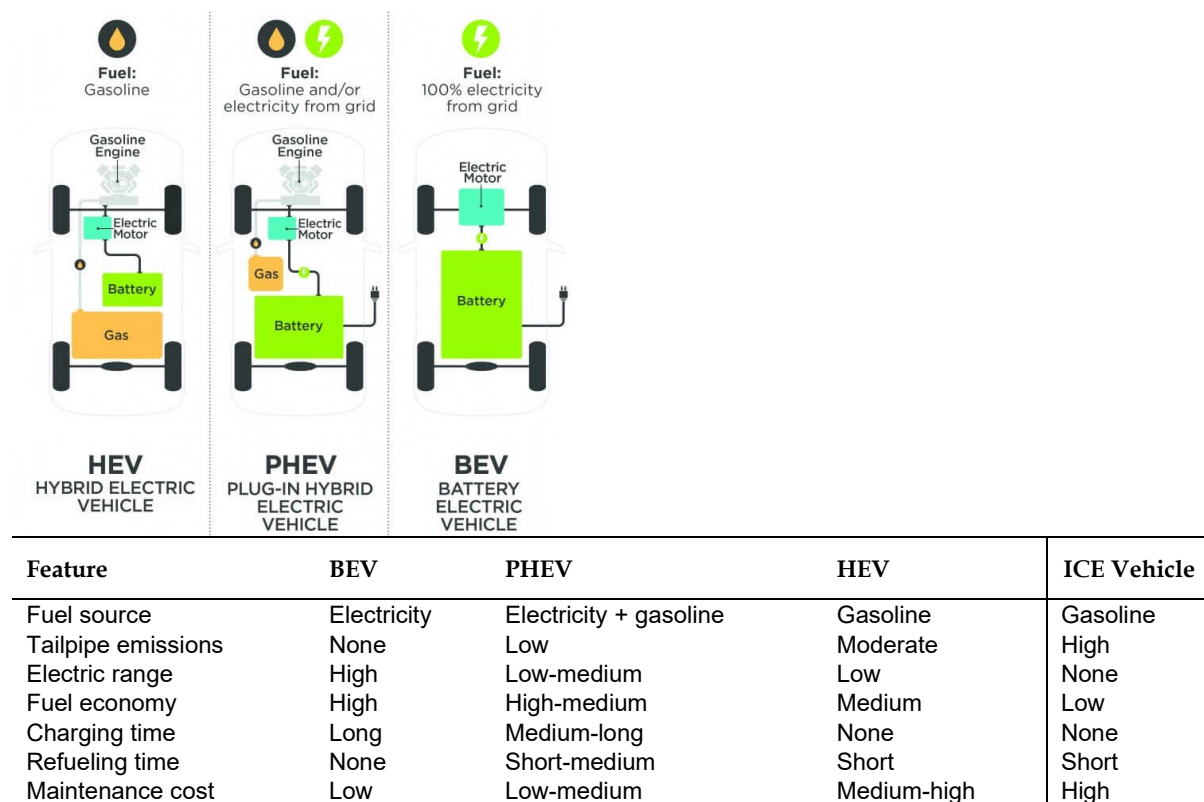
Source: IEA, Price of selected battery materials and lithium-ion batteries, 2015-2023, (Paris: International Energy Agency, 2023), License: CC BY 4.0, <https://tinyurl.com/2p88yph>, accessed August 2023.

Note: Dots indicate Li-ion battery prices, indexed on the right axis. Lines indicate metal prices, indexed on the left axis.

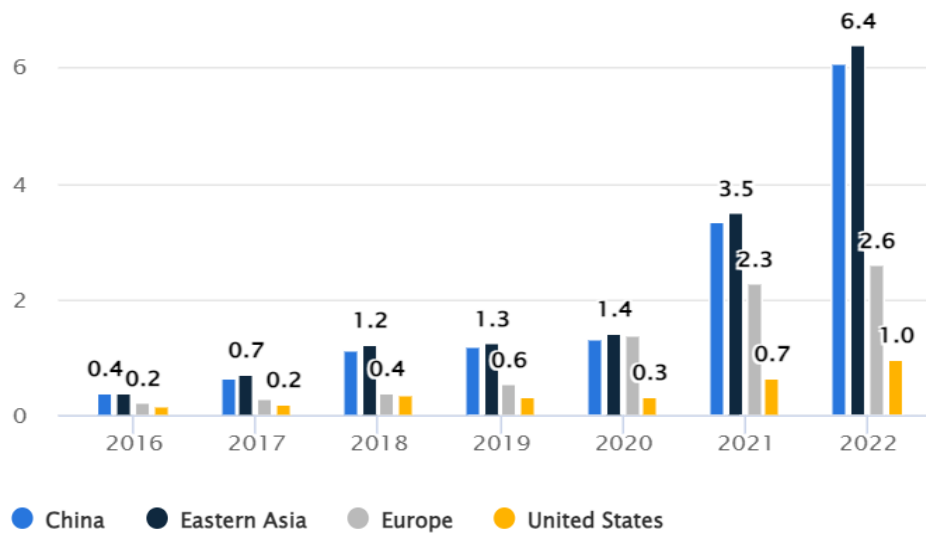
Exhibit 5 Chinese EV Market Share, 2022 (in %)

Company	Share of EV Market
BYD	30%
Wuling	7%
Tesla	7%
Aion	4%
Changan	4%
Chery	4%
Volkswagen	3%
Others	43%

Source: Howard Lai, "China's Electric Vehicle Market, a Rising Global Leader in EV Technology," Daxue Consulting, July 17, 2023, <https://daxueconsulting.com/electric-vehicle-market-in-china/>, accessed July 2023.

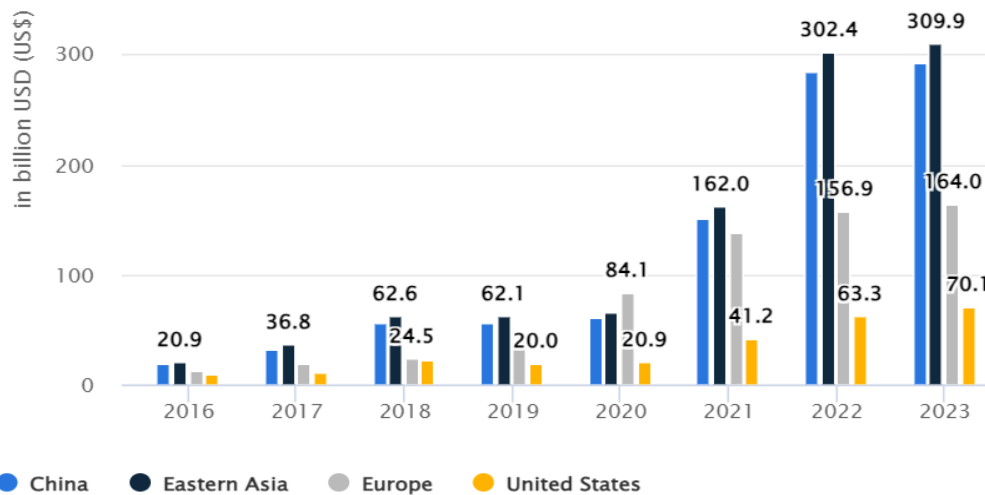
Exhibit 6 Electric Vehicle Types and Characteristics

Source: U.S. Department of Energy, "All-Electric Vehicles," Alternative Fuels Data Center, <https://bit.ly/47zzLau>; Electric Power Research Institute, cited in Auto Outlet USA, "What Are the Different Types Of Electric Vehicles," January 12, 2021, <https://bit.ly/3Wmbu2a>, both accessed August 2023.

Exhibit 7a EVs Sold by Region/Country, 2016-2022 (in million units)

Source: "Electric Vehicles," via Statista Outlook, May 2023, <https://tinyurl.com/49c94bby>, accessed August 2023.

Note: EVs include battery electric and plug-in hybrid vehicles.

Exhibit 7b EV Revenue by Region/Country, 2016-2023 (in \$ billions)

Source: "Electric Vehicles," via Statista Outlook, May 2023, <https://tinyurl.com/49c94bby>, accessed August 2023.

Note: 2023 results are estimates. EVs include battery electric and plug-in hybrid vehicles.

Exhibit 8a Share of EV Sales by Make, 2022 (in percent)

China		East Asia		Europe		United States	
BYD	29.6%	BYD	28.2%	Tesla	8.6%	Tesla	51.8%
Wuling	7.2	Tesla	7.4	BMW	8.3	Ford	7.0
Tesla	7.1	Wuling	6.9	Mercedes	8.3	Jeep	5.1
Aion	4.4	Aion	4.2	Volkswagen	8.3	BMW	4.5
Changan	3.8	Changan	3.6	Audi	5.4	Chevrolet	3.8
Chery	3.6	Chery	3.5	Kia	5.2	Kia	3.8
Volkswagen	3.1	Volkswagen	3.0	Peugeot	5.2	Hyundai	3.4
Hozon Auto	2.4	Hozon Auto	2.3	Volvo	5.1	Toyota	3.2
Li Xiang	2.2	Li Xiang	2.1	Renault	4.6	Volvo	2.8
Nio	2.0	Nio	1.9	Hyundai	4.5	Volkswagen	2.1
Others	34.5	Others	36.9	Others	36.6	Others	12.5

Source: "Electric Vehicles – Worldwide," via Statista Outlook, May 2023, <https://tinyurl.com/49c94bby>, accessed August 2023.

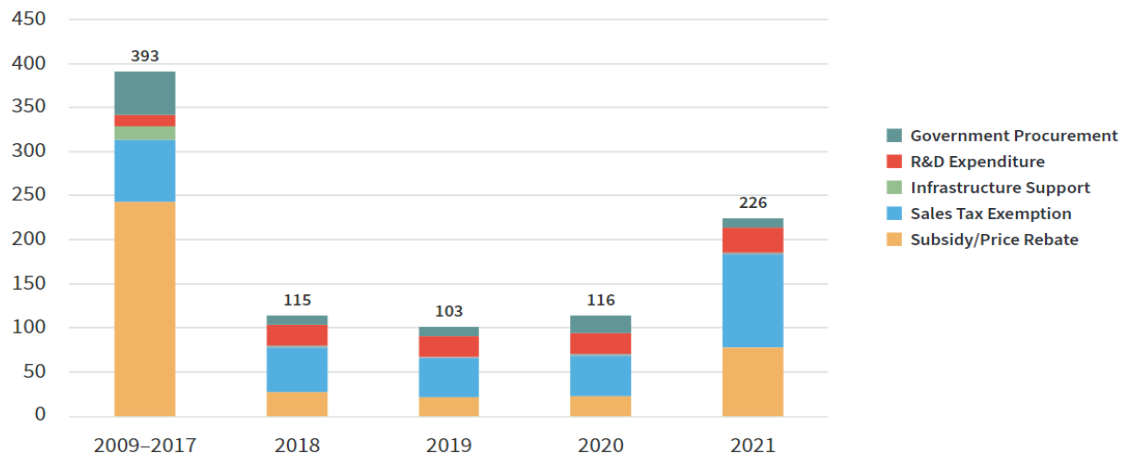
Note: East Asia includes China, Japan, and South Korea.

Exhibit 8b Share of EV Revenue by Make, 2021 (in percent)

China		East Asia		Europe		United States	
Tesla	23.9%	Tesla	24.0%	Mercedes	11.6%	Tesla	65.2%
BYD	13.0	BYD	11.9	Tesla	10.5	Jeep	4.8
Wuling	6.0	Wuling	5.5	BMW	9.9	Toyota	4.0
Nio	5.5	Nio	5.0	Volkswagen	8.8	Volvo	4.0
Li Xiang	5.3	Li Xiang	4.8	Audi	7.9	BMW	3.3
Volkswagen	3.8	BMW	3.6	Volvo	6.8	Ford	2.9
Cowin	3.4	Volkswagen	3.5	Kia	4.3	Audi	2.8
Roewe	3.3	Cowin	3.1	Peugeot	4.2	Chevrolet	2.0
Porsche	2.8	Porsche	3.0	Renault	4.1	Porsche	2.0
BMW	2.7	Roewe	3.0	Hyundai	3.3	Volkswagen	1.4
Other	30.3	Other	32.7	Other	28.6	Other	7.7

Source: "Electric Vehicles – Worldwide," via Statista Outlook, May 2023, <https://tinyurl.com/49c94bby>, accessed August 2023.

Note: East Asia includes China, Japan, and South Korea.

Exhibit 9 Chinese Government Spending on New Energy Vehicles, 2009-2021 (in RMB billions)

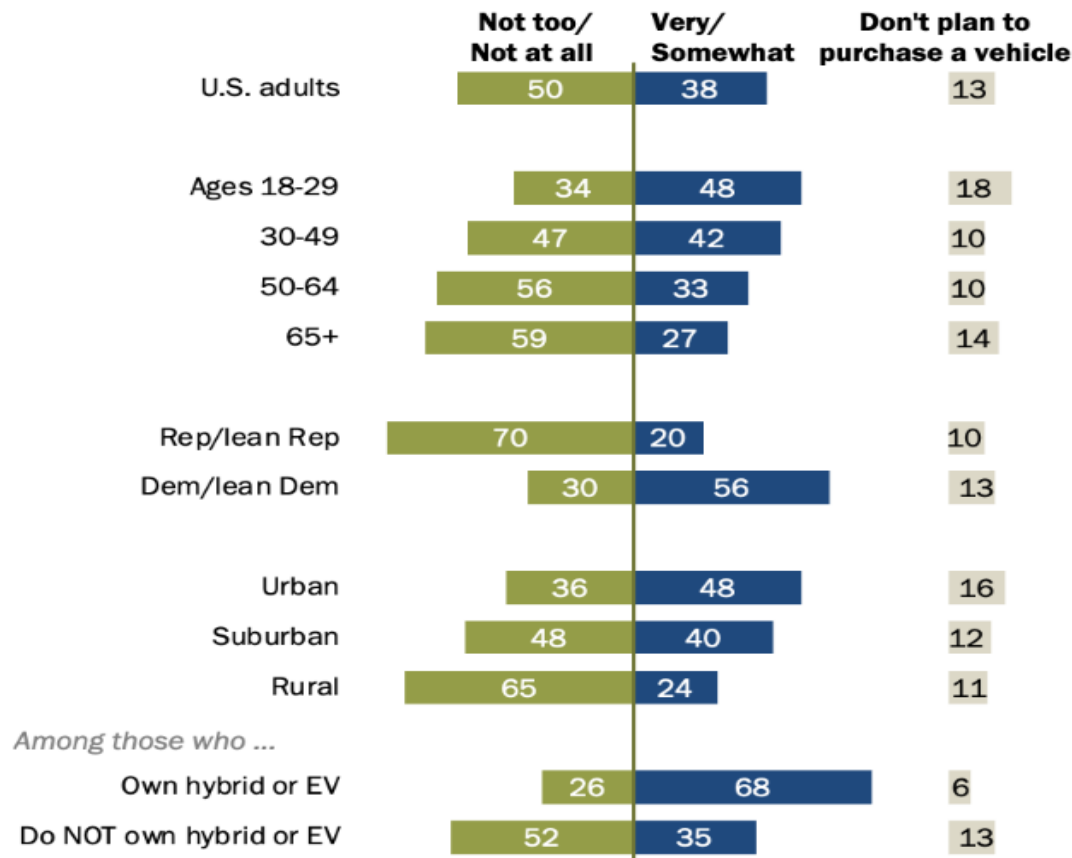
Source: Gerard DiPippo, Ilaria Mazzocco, and Scott Kennedy, "Red Ink: Estimating Chinese Industrial Policy Spending in Comparative Perspective," The Center for Strategic and International Studies, May 2022, p. 56, <https://tinyurl.com/57wz745j>, accessed August 2023.

Note: Using average annual exchange rates, these figures amount to \$58 billion for the 2009-2017 period, \$17 billion in 2018, \$15 billion in 2019, \$17 billion in 2020, and \$35 billion in 2021, yielding a total of \$144 billion between 2009 and 2021, and an unstated \$29 billion in 2022 spending. Casewriter calculation from Macrotrends, "China-U.S. Foreign Exchange Rate—Historical Annual Data," <https://tinyurl.com/2s459yw2>, accessed August 2023.

Exhibit 10 Charging Locations per 100 km of Roadway by Country, 2021

Country	Charging locations per 100 km of roadway	Total length of road network, km
South Korea	75.2	100,428
Netherlands	21.9	139,124
Norway	13.6	94,902
China	9.8	5,200,000
Belgium	5.6	118,414
Sweden	5.2	573,134
UK	4.7	394,428
United Arab Emirates	2.7	4,080
Germany	2.7	830,000
Singapore	2.6	3,500
Israel	2.3	20,391
Japan	1.7	1,218,772
France	1.6	1,053,215
Spain	1.4	683,175
Italy	1.0	487,700
United States	0.8	6,586,610
Canada	0.8	1,042,300

Source: "Charging locations per 100 kilometers of roadway in 2021, by country," October 2021, via Statista; CIA World Factbook, "Field Listing—Roadways," www.cia.gov/the-world-factbook/field/roadways/, accessed August 2023.

Exhibit 11 U.S. Consumer EV Purchasing Preferences, 2023

Source: Alison Spencer, Stephanie Ross, and Alec Tyson, "How Americans View Electric Vehicles," Pew Research Center, July 13, 2023, <https://tinyurl.com/mwm5mcbd>, accessed July 2023.

Note: Exhibit shows the percentage of U.S. adults who say they are ____ likely to seriously consider purchasing an electric vehicle the next time they purchase a vehicle.

Appendix A Powering EVs

Types of EVs

Rather than rely on internal combustion engines (ICEs), EVs used motors powered by electricity from various sources, such as batteries, fuel cells, or the electric grid. Three main types of EVs were battery electric vehicles (BEVs), plug-in hybrid electric vehicles (PHEVs), and hybrid electric vehicles (HEVs). They differed in their inclusion or elimination of supplemental ICE support. BEVs used only batteries to store electricity, had no other fuel source, were charged by plugging into an external power outlet or a charging station, and had no tailpipe emissions. PHEVs used both batteries and another fuel source, usually gasoline, to power an ICE or a generator. They ran on electricity alone for a limited distance, usually between 15 to 60 miles, before switching to the ICE or generator. Like BEVs, PHEVs relied on an external power outlet or a charging station for power, augmented by regenerative braking and an ICE or generator. HEVs also used both batteries and another fuel source, usually gasoline, to power an ICE or a generator, but they could not run on electricity alone and had no plug-in capability. HEVs were charged by regenerative braking and the ICE or generator and had higher fuel consumption and tailpipe emissions than other EVs, yet were still lower than ICE vehicles.

Batteries

Lithium-ion (Li-ion), nickel-metal hydride, and lead-acid batteries were the main types of energy storage system for EVs, along with ultracapacitors. Nickel-metal hydride batteries lasted longer and were safer than lead-acid batteries but were expensive, lost charge quickly, and got hot; they also needed to be sealed to prevent hydrogen from escaping. Lead-acid batteries were cheap and reliable, with low energy and power; they performed poorly in cold weather and wore out quickly, so they were typically only used in EVs to carry extra load (e.g., to boost the audio/video or climate control systems). Ultracapacitors provided extra power for acceleration and hills.

Li-ion batteries were the most commonly used because they stored more energy per unit of mass than other systems, with high power, efficiency, and temperature performance and low self-discharge (i.e., kept their charge a long time while unused). However, their disposal and recycling were expensive and difficult.

In 2022, demand for Li-ion batteries rose by 65% from 2021 based on a 55% increase in passenger EV registrations. China led the way, with EV sales up 80% and demand for Li-ion batteries up by over 70% compared to 2021. By 2023, China produced some 60% of the world's EV batteries. In the U.S., demand for Li-ion batteries rose 80% despite EV sales increasing by just 55%.

Demand increases drove supply shortages of critical materials. Worldwide, in 2022, 60% of lithium, 30% of cobalt, and 10% of nickel went into EV batteries, compared to 15%, 10%, and 2% in 2017, respectively. From 2017 to 2022, lithium production increased 180%, yet demand outstripped supply in 2021 and 2022. Although mineral extraction occurred worldwide, in 2023, Chinese companies refined 70% of the world's cobalt, nearly 60% of all lithium, and almost 25% of nickel ores.

In 2022, Li-ion battery alternatives were becoming more popular, including nickel manganese cobalt oxide (NMC), lithium iron phosphate (LFP), and nickel cobalt aluminum oxide (NCA) technology. Again, China led the trend: 50% of all LFP batteries for passenger vehicles were for BYD, and 95% were for vehicles manufactured in China. Tesla purchased 15% of LFP batteries for EVs worldwide (using 85% of its purchases for vehicles manufactured in China); 3% of LFP EVs were made in the U.S.

Minerals including cobalt, nickel, and lithium were used to make EV batteries. Congo, Indonesia, and South America's Southern Cone were rich in these minerals. China mined around 90% of all rare-earth elements, including 60% of materials used in the complete stages of EV battery production. The price and availability of these critical materials influenced battery trends. Cobalt was expensive and controversial (often mined in poor and unsafe conditions, sometimes by children, and could cause health and environmental problems), so battery makers reduced its use in NMC batteries or switched to LFP batteries, which used different materials and were cheaper and safer, although they had lower energy density. Lithium was also expensive and scarce, and Chinese battery makers were starting to adopt sodium-ion batteries as a cheap alternative that did not need these critical minerals. The demand for critical minerals was higher than the supply in 2021 and 2022, driving up their prices.

Resource-exporting countries entered talks to control the supply of lithium and other minerals. Lithium-rich nations in South America's Southern Cone discussed the possibility of a "lithium OPEC" (see note). The CEO of French carmaker Renault admitted that the prospect of Chinese policymakers cutting off U.S. and European firms from Chinese minerals kept him up at night: "China – and no one can blame them for that – is putting its hand on mines and especially on the transformation of metals used to build batteries. The war of the future will be a war of metals."

BYD Seeks Materials for Batteries

From 2009 to 2021, the Chinese government pumped \$125 billion in investment into its domestic EV industry and supply chain. China also invested in nickel and lithium sources abroad. Indonesian business leaders and policymakers lobbied their government to allow China to build factories in Indonesia, moreover, and in 2023, BYD entered talks with the Indonesian government to build an "EV ecosystem" in the country. That year, VW also announced plans to partner with Ford to open an Indonesian battery production facility.

In April 2023, BYD announced plans to open a \$290 million lithium cathode factory in northern Chile. The factory was slated to produce 50,000 tons of lithium iron phosphate every year, beginning in 2025; as a designated "qualified lithium producer," BYD would receive preferential pricing on its Chilean lithium precursor materials. Said a Chilean economic development official, "We hope to give a strong boost to promote the generation of products in the lithium value chain."

Source: Compiled by casewriter from U.S. Department of Energy, "Plug-In Hybrids," Alternative Fuels Data Center, <https://bit.ly/47zzLau>; U.S. Department of Energy, "Batteries for Electric Vehicles," Alternative Fuels Data Center, <https://bit.ly/456F3bX>; IEA, Global EV Outlook 2023, (Paris: International Energy Agency, 2023), License: CC BY 4.0, <https://www.iea.org/reports/global-ev-outlook-2023>; Edward White, "How China Cornered the Market for Clean Tech," Financial Times, August 9, 2023, <https://on.ft.com/3YEI9BJ>; Jordan Brinn, "Electric Vehicle Supply Chains: The Basics," Natural Resources Defense Council, July 7, 2022, <https://tinyurl.com/4crerzmk>; Kate Mackenzie and Tim Sahay, "Mercantilist Deals of the Great Powers," Phenomenal World, April 6, 2023, <https://tinyurl.com/mrxhae7m>; Ania Nussbaum, "Renault Chairman Says China Raw Material Risk Keeps Him Up at Night," Bloomberg, July 8, 2023, <https://tinyurl.com/2amfhank>; Daniel Ren, "BYD Signs Agreement to Explore EV Opportunities in Indonesia," *South China Morning Post*, May 29, 2023, <https://tinyurl.com/4b6t656c>; "Volkswagen to partner on Indonesia EV battery ecosystem minister," Reuters, April 17, 2023, <https://tinyurl.com/3m5u5wt5>; "China EV Maker BYD to Build \$290 Million Battery Component Plant in Chile," Reuters, April 20, 2023, <https://tinyurl.com/3wnhan3c>; all accessed August 2023.

Note: In 1960, the governments of Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela founded OPEC, the Organization of the Petroleum Exporting Countries, to set oil price levels and production targets.

Appendix B Charging Electric Vehicles

Governments played a role in supporting the supply of EV charging stations. Consumer anxiety over running out of battery power while traveling was a leading reason for reluctance to consider purchasing an EV; areas that had more chargers saw a corresponding increase in EV purchases.

Charging stations could be fast or slow, the former for on-the-go charging in public areas and the latter for overnight charging at home. In the U.S., “Level 1” slow chargers provided power at roughly 5 miles of range per hour of charging, while “Level 2” slow chargers provided 25 miles of range per hour. Fast chargers provided 100 to 200 miles of range or more per 30 minutes of charging, and were mostly used in public, high-traffic areas such as alongside highways. Areas with many single-family homes—which were more likely to be able to install a home charger—often required and constructed fewer stations than those where drivers relied mostly on public chargers. Areas with more fast chargers needed fewer overall chargers, since fast chargers could serve more EVs in a day than slow chargers.

U.S. By 2023, the U.S. had 28,000 fast chargers and 24 EVs for every public charger.

While General Motors promoted an electric car in the 1990s, a combination of low driving ranges, government indifference, and oil company lobbying were blamed for scuttling the project and, some claimed in the 2006 documentary *Who Killed the Electric Car?*, the entire American EV industry. In 2009, U.S. President Barack Obama took office and spent eight years bolstering the EV industry with investments in research and rebates for consumers; his administration provided \$4.5 billion in incentives to construct charging stations and designated “electric vehicle charging corridors” where drivers could find a charger every 50 miles or less. From 2011 to 2013, the number of EV charging stations in the country grew from 2,100 to 6,200; by 2020 there were 32,000 stations.

In 2022, as part of a broader infrastructure funding bill, the U.S. launched the \$5 billion National Electric Vehicle Infrastructure initiative and a \$2.5 billion Charging and Fueling Infrastructure grant program, both contributing to the goal of building 500,000 EV chargers by 2030. In February 2023, the U.S. announced new rules for standardization of connectors and other aspects of charging stations.

Europe By 2023, Europe had 460,000 slow chargers and more than 70,000 fast chargers. It had 13 EVs for every public charger. Germany had 64,000 slow chargers, 12,000 fast chargers, and 26 EVs per charger; France had 74,000 slow chargers, 9,700 fast chargers, and 13 EVs per charger; and the Netherlands had 117,000 slow chargers and four EVs per charger.

That year, the EU passed a law requiring its member countries ensure installation of fast chargers every 37 miles on the main highway system; it also required payment to be accepted via card to preclude operators from locking drivers into downloading an app or signing up for a subscription. In addition to EU-wide incentives such as a €1.5 billion annual grant program that aimed to construct 3.5 million chargers by 2030, many EU countries had their own incentive programs; these national and EU-wide policies supplemented each other. To meet its goal of 1 million charging stations and 10 million EVs by 2030, for example, Germany offered businesses up to €45,000 per location to install a charging station, and residents €900 to install a home charger; €350 million was allocated for 2022.

China By 2023, China had more than 1 million slow chargers, 760,000 fast chargers, and was expanding quickly—90% of the 330,000 fast chargers built in 2022 were in China. It had eight EVs for every public charger.

The Chinese government began planning an EV industry in the 1990s. In 2013, to remedy a patchwork of different charging stations, it mandated interoperability—even Tesla Superchargers in

the country were compatible with all other chargers. The government mandated that all parking spaces in new residential buildings contain chargers. Because of China's strong national rail service, most EV chargers were focused on travel within cities rather than by highway – 70% of public EV chargers were located in dense urban areas.

Source: Compiled by casewriter from Sasha Lekach and Alex Kwanten, "EV Survey Shows Range Anxiety High While Charging Costs Remain Low," *Forbes*, June 28, 2022, <https://tinyurl.com/4raac2ja>; Alternative Data Fuels Center, "Developing Infrastructure to Charge Electric Vehicles," U.S. Department of Energy, <https://tinyurl.com/2rstmea>; IEA, "Global EV Outlook 2023: Trends in Charging Infrastructure," April 2023, <https://tinyurl.com/yxk9b5p9>; James Morris, "Why Nobody Will Kill the Electric Car This Time," *Forbes*, February 20, 2021, <https://tinyurl.com/4ckz8r6w>; Alex Davies, "Obama Caps Off His EV Legacy by Electrifying American Highways," *Wired*, November 4, 2016, <https://tinyurl.com/ypx88sbt>; Bureau of Transportation Statistics, "Number of Alternative Fueling Stations Nearly Doubles since 2020; EV Charging Leads the Way," U.S. Department of Transportation, April 17, 2023, <https://tinyurl.com/de2pnrdrn>; The White House, "FACT SHEET: Biden-Harris Administration Announces New Standards and Major Progress for a Made-in-America National Network of Electric Vehicle Chargers," press release, February 15, 2023, <https://tinyurl.com/2e9af49f>; Thomas Ricker, "EU Passes Law to Blanket Highways with Fast EV Chargers by End of 2025," *The Verge*, July 25, 2023, <https://tinyurl.com/32s5pnb3>; EVBox, "EV charging infrastructure incentives in Europe 2022," April 22, 2022, <https://tinyurl.com/yw8ftjpt>; Jennifer Conrad, "China is Racing to Electrify its Future," *Wired*, June 29, 2022, <https://tinyurl.com/2b3hd3uu>; all accessed August 2023.

Endnotes

- ¹ James E. Ellis, "BYD Is Racing to the Top of the Global EV Market," *Bloomberg Business Week*, June 12, 2023, accessed July 2023.
- ² Meaghan Tobin and Lyric Li, "Chinese EV Giant BYD Overtakes Tesla, but Can It Crack the U.S. Market?" *The Washington Post*, June 14, 2023, <https://tinyurl.com/ycy6jehy>, accessed July 2023.
- ³ Rita Liao, "BYD Is Overtaking Tesla, but Its EV Dreams Skip the US for Now," *TechCrunch*, June 15, 2023, <https://techcrunch.com/2023/06/15/byd-is-overtaking-tesla-but-its-ev-dream-skips-the-us-for-now/>, accessed July 2023.
- ⁴ Liao, "BYD is overtaking Tesla, but its EV dreams skip the US for now."
- ⁵ Tobin and Li, "Chinese EV Giant BYD Overtakes Tesla, but Can It Crack the U.S. Market?"
- ⁶ Tobin and Li, "Chinese EV Giant BYD Overtakes Tesla, but Can It Crack the U.S. Market?"
- ⁷ Amy Hawkins, "Battery Power: How China Could Take Control of the Electric Vehicle Market," *The Guardian*, July 29, 2023, <https://tinyurl.com/2vers66n>, accessed July 2023.
- ⁸ Howard Lai, "China's Electric Vehicle Market, a Rising Global Leader in EV Technology," *Daxue Consulting*, July 17, 2023, <https://daxueconsulting.com/electric-vehicle-market-in-china/>, accessed July 2023.
- ⁹ Umesh Bhagchandani, "Forget Elon Musk, Meet Wang Chuanfu: the Chinese Billionaire Has Dethroned Tesla with BYD's 2022 Electric Car Sales and Was Backed by Warren Buffet—but Comes from a Family of Farmers," *South China Morning Post*, August 9, 2022, <https://tinyurl.com/473uubzr>, accessed August 2023.
- ¹⁰ Daniel Zlatev, "Tesla Price Cut Comes as BYD Calls for Global EV Domination," *Notebookcheck*, August 14, 2023, <https://tinyurl.com/bddhtna9>, accessed August 2023.
- ¹¹ Zlatev, "Tesla Price Cut Comes as BYD Calls for Global EV Domination."
- ¹² Gloria Li and Edward White, "Only 10 Carmakers Will Survive Global EV Battle, Says Tesla Rival XPeng," *Financial Times*, April 24, 2023, <https://www.ft.com/content/852fd11c-46dc-4d14-8254-37aa360470f7>, accessed July 2023.
- ¹³ Lai, "China's Electric Vehicle Market, a Rising Global Leader in EV Technology."
- ¹⁴ Reuters, "Battle for China's Electric SUV Market Heats Up at Home and Abroad," May 10, 2023, <https://tinyurl.com/3a9v9jah>, accessed July 2023.
- ¹⁵ Lai, "China's Electric Vehicle Market, a Rising Global Leader in EV Technology."
- ¹⁶ "Electric Vehicles – Worldwide," via Statista Outlook, May 2023, <https://tinyurl.com/49c94bby>, accessed August 2023.
- ¹⁷ Reuters, "Battle for China's Electric SUV Market Heats Up at Home and Abroad."
- ¹⁸ Greg Ip, "China's EV Juggernaut Is a Warning to the West," June 7, 2023, <https://tinyurl.com/3dtsa2mw>, accessed July 2023.
- ¹⁹ Zlatev, "Tesla Price Cut Comes as BYD Calls for Global EV Domination."
- ²⁰ Li and White, "Only 10 Carmakers Will Survive Global EV Battle, Says Tesla Rival XPeng."
- ²¹ Melissa Eddy, "China's Leading Electric Carmaker Has Arrived in Germany," *The New York Times*, February 21, 2023, <https://www.nytimes.com/2023/02/21/world/europe/byd-germany-electric-cars.html>, accessed July 2023.
- ²² Eddy, "China's Leading Electric Carmaker Has Arrived in Germany."
- ²³ Reuters, "Tesla Offers New China Rebate as Price Cuts Rock EV Market," July 7, 2023, <https://tinyurl.com/5a5u2xft>, accessed July 2023.
- ²⁴ Reuters, "Tesla Offers New China Rebate as Price Cuts Rock EV Market."
- ²⁵ Reuters, "Battle for China's Electric SUV Market Heats Up at Home and Abroad."
- ²⁶ Reuters, "Battle for China's Electric SUV Market Heats Up at Home and Abroad."

- ²⁷ Joey Klender, "Tesla Price Cuts in China Drop Competitor Sales by as Much as 60%," *Teslarati*, February 1, 2023, <https://tinyurl.com/6rvx2f8y>, accessed August 2023.
- ²⁸ Elise Mak, "Chinese NEV Makers Need to Offer Better Value Proposition to Buyers as Price War 'Unsustainable in the Long-Term', Says Fitch," *South China Morning Post*, May 19, 2023, <https://tinyurl.com/bdszkbz>, accessed August 2023.
- ²⁹ Mak, "Chinese NEV Makers Need to Offer Better Value Proposition to Buyers as Price War 'Unsustainable in the Long-Term', Says Fitch."
- ³⁰ Leonardo Lara, "BYD Picks Brazil for Its First Electric-Vehicle Hub outside Asia," *Bloomberg*, July 4, 2023, <https://tinyurl.com/35azdmf2>, accessed July 2023.
- ³¹ Kate Mackenzie and Tim Sahay, "Mercantilist Deals of the Great Powers," *Phenomenal World*, April 6, 2023, <https://tinyurl.com/mrxhae7m>, accessed July 2023.
- ³² Ellis, "BYD Is Racing to the Top of the Global EV Market."
- ³³ Sean McLain, "How Did Hyundai Get So Cool?" *The Wall Street Journal*, May 22, 2023, <https://tinyurl.com/3vrks9y9>, accessed August 2023.
- ³⁴ Ellis, "BYD Is Racing to the Top of the Global EV Market;" and "Lu Xiangyang," *Forbes Profile*, August 21, 2023, <https://tinyurl.com/yc25s244>; both accessed July 2023.
- ³⁵ Che Pan, "BYD Scraps Spin-Off of Chip Unit, Citing New Investment and Restructuring to Tap Electric Vehicle Market," *South China Morning Post*, November 16, 2022, <https://tinyurl.com/3j4hw9u8>, accessed July 2023.
- ³⁶ Tycho de Feijter, "This History of the Flyer – From Qinchuan Alto to BYD," *China Car History*, November 4, 2020, <https://chinacarhistory.com/2020/04/11/this-history-of-the-flyer-from-qinchuan-alto-to-byd/>, accessed August 2023.
- ³⁷ Ellis, "BYD Is Racing to the Top of the Global EV Market."
- ³⁸ Joan Muller, "Thanks, Now Move Over," *Forbes*, July 26, 2004, <https://tinyurl.com/ysc4mks4>, accessed August 2023.
- ³⁹ Eddy, "China's Leading Electric Carmaker Has Arrived in Germany."
- ⁴⁰ Ellis, "BYD Is Racing to the Top of the Global EV Market."
- ⁴¹ "The Tech 100," *Bloomberg Business Week*, May 24, 2010, <https://tinyurl.com/y23489nz>, accessed August 2023.
- ⁴² Ellis, "BYD Is Racing to the Top of the Global EV Market."
- ⁴³ Ellis, "BYD Is Racing to the Top of the Global EV Market."
- ⁴⁴ Ellis, "BYD Is Racing to the Top of the Global EV Market."
- ⁴⁵ Ellis, "BYD Is Racing to the Top of the Global EV Market."
- ⁴⁶ Ellis, "BYD Is Racing to the Top of the Global EV Market."
- ⁴⁷ Eddy, "China's Leading Electric Carmaker Has Arrived in Germany."
- ⁴⁸ Ellis, "BYD Is Racing to the Top of the Global EV Market;" and Market cap: Companies Market Cap, "Market Cap History of BYD from 2011 to 2023," August 10, 2023, <https://tinyurl.com/ywuarn6m>; both accessed August 2023.
- ⁴⁹ Battery Bits Editors, "BYD: How the Former EV Pioneer Will Win the New Battle," *Medium*, April 25, 2021, <https://tinyurl.com/463e4duc>, accessed August 2023.
- ⁵⁰ Russell Flannery, "Plugged In: BYD's Wang Chuanfu Explains How China's No. 1 EV Maker Caught Up with Tesla," *Forbes*, November 9, 2022, <https://tinyurl.com/48n84np4>, accessed August 2023.
- ⁵¹ Bruce Einhorn and Danny Lee, "Chinese EV Maker BYD Aims to Conquer World Markets as the Un-Tesla," *Asia Pacific Metalworking Equipment News*, August 19, 2022, <https://tinyurl.com/2z4bymc4>, accessed August 2023.
- ⁵² Norihiko Shirouzou, "Focus: China's BYD Takes Cautious Approach to U.S. in Global EV Push," *Reuters*, January 18, 2023, <https://tinyurl.com/2a5m4xz8> accessed July 2023.
- ⁵³ Shirouzou, "Focus: China's BYD Takes Cautious Approach to U.S. in Global EV Push."

- ⁵⁴ Shirouzou, "Focus: China's BYD Takes Cautious Approach to U.S. in Global EV Push."
- ⁵⁵ Shirouzou, "Focus: China's BYD Takes Cautious Approach to U.S. in Global EV Push."
- ⁵⁶ Joey Fung, "China's Carmakers BYD and SAIC Are Avoiding the Supply Chain Crisis by Ordering Their Own Ships," *The Millennial Source*, December 6, 2022, <https://tinyurl.com/2p8xe3a5>, accessed August 2023.
- ⁵⁷ Edward White, "How China Cornered the Market for Clean Tech," *Financial Times*, August 9, 2023, <https://on.ft.com/3YEI9BJ>, accessed August 2023.
- ⁵⁸ Yusuke Hinata, "BYD to Start Selling In-Vehicle Batteries to Other Companies," *Nikkei Asia*, April 3, 2021, <https://tinyurl.com/y98bpny7>, accessed August 2023.
- ⁵⁹ Frost & Sullivan, "Strategic Profile of BYD," December 12, 2022, <https://tinyurl.com/2w2nyb6u>, accessed August 2023.
- ⁶⁰ Monika, "BYD Scales Up Battery Production Capacity to Support Fast-Growing NEV Sales," *Gasgoo*, December 13, 2021, <https://autonews.gasgoo.com/m/70019325.html>, accessed July 2023.
- ⁶¹ Daniel Ren, "China's BYD Breaks Ground on EV Battery Plant, Betting Nearly US\$1 Billion despite Concerns of Oversupply," *South China Morning Post*, February 28, 2023, <https://tinyurl.com/mrykfdyk>, accessed July 2023.
- ⁶² Ren, "China's BYD Breaks Ground on EV Battery Plant, Betting Nearly US\$1 Billion despite Concerns of Oversupply."
- ⁶³ Ren, "China's BYD Breaks Ground on EV Battery Plant, Betting Nearly US\$1 Billion despite Concerns of Oversupply."
- ⁶⁴ Ren, "China's BYD Breaks Ground on EV Battery Plant, Betting Nearly US\$1 Billion despite Concerns of Oversupply."
- ⁶⁵ Hawkins, "Battery Power: How China Could Take Control of the Electric Vehicle Market;" Adam Ismail, "Toyota Can't Do This Alone Anymore," *Jalopnik*, December 3, 2021, <https://tinyurl.com/4ywwfxtn>; José Rodríguez Jr., "The Absurdly Cheap BYD Seagull Could Become China's Bestselling EV," *Jalopnik*, April 19, 2023, <https://tinyurl.com/4dkxzzcr>; all accessed August 2023.
- ⁶⁶ Phate Zhang, "BYD Semiconductor to Put Chip Production Line in Hunan into Operation in October," *CnEVPPost*, September 13, 2022, <https://tinyurl.com/9ucbmchh>, accessed July 2023.
- ⁶⁷ Mike Marut, "Inside a Semiconductor Chip Fab: Austin Makes Some of the World's Most Desired Products," *KVUE*, May 31, 2022, <https://tinyurl.com/2e8vssvp>, accessed August 2023.
- ⁶⁸ Reuters, "China's BYD Scraps IPO Plan for Semiconductor Unit," November 15, 2022, <https://tinyurl.com/bddxtudk>, accessed July 2023.
- ⁶⁹ Reuters, "China's BYD Scraps IPO Plan for Semiconductor Unit."
- ⁷⁰ Yusuke Hinata, "BYD Eyes Overtaking Tesla This Year as World's Top EV Maker," *Nikkei Asia*, April 7, 2023, <https://tinyurl.com/547j65ws>, accessed July 2023.
- ⁷¹ Mark Kane, "Tesla Production Sites by Model Assignment, Capacity: April 2023," *Electrek*, April 2023, <https://tinyurl.com/yuv7v3zt>, accessed August 2023.
- ⁷² Kane, "Tesla Production Sites by Model Assignment, Capacity: April 2023."
- ⁷³ Hinata, "BYD Eyes Overtaking Tesla This Year as World's Top EV Maker."
- ⁷⁴ International Energy Agency (IEA), "Global EV Outlook: Executive Summary," April 2023, <https://tinyurl.com/4w4bb9sv>, accessed July 2023.
- ⁷⁵ IEA, "Global EV Outlook: Executive Summary."
- ⁷⁶ "Electric Vehicles, A Global Overview," via Statista, p. 8, <https://tinyurl.com/mr44ystk>, accessed August 2023.
- ⁷⁷ Rebecca Matulka, "The History of the Electric Car," U.S. Department of Energy, September 15, 2014, <https://www.energy.gov/articles/history-electric-car>, accessed August 2023.
- ⁷⁸ Matulka, "The History of the Electric Car."
- ⁷⁹ Matulka, "The History of the Electric Car."
- ⁸⁰ Matulka, "The History of the Electric Car."

- ⁸¹ Matulka, "The History of the Electric Car."
- ⁸² "Electric Vehicles – Worldwide," via Statista Outlook, May 2023, <https://tinyurl.com/49c94bby>, accessed August 2023.
- ⁸³ Marcus Lu, "Visualizing EV Production in the U.S. by Brand," Visual Capitalist, December 22, 2022, <https://www.visualcapitalist.com/ev-production-by-brand-united-states/>, accessed August 2023.
- ⁸⁴ Lu, "Visualizing EV Production in the U.S. by Brand."
- ⁸⁵ Fully Charged, "Electric Vehicle Database: Range of Full Electric Vehicles," 2023, <https://tinyurl.com/w35ftcv5>, accessed August 2023.
- ⁸⁶ Renee Valdes, "How Long Does It Take to Charge an Electric Car?" Kelley Blue Book, July 11, 2023, <https://www.kbb.com/car-advice/how-long-does-it-take-to-charge-an-electric-car/>, accessed August 2023.
- ⁸⁷ Huileng Tan, "The Chinese EV Market Is Exploding. Here Are the 5 Major Chinese EV Brands You Should Know About," Business Insider, May 19, 2023, <https://tinyurl.com/mstb5drd>, accessed August 2023.
- ⁸⁸ Jameson Dow, "NIO's EP9 Beats Every Production Car Ever with a Nurburgring Lap Record of 6:45.90," Electrek, May 13, 2017, <https://tinyurl.com/yckm37mn>, accessed August 2023.
- ⁸⁹ Tan, "The Chinese EV Market Is Exploding. Here Are the 5 Major Chinese EV Brands You Should Know About."
- ⁹⁰ Tan, "The Chinese EV Market Is Exploding. Here Are the 5 Major Chinese EV Brands You Should Know About."
- ⁹¹ Keith Bradsher, "In China, a Big Auto Show Returns to a Country That Has Gone Electric," *The New York Times*, April 19, 2023, <https://www.nytimes.com/2023/04/19/business/shanghai-auto-show-electric-vehicles.html>, accessed August 2023.
- ⁹² Jack Ewing, "In Norway, the Electric Vehicle Future Has Already Arrived," *The New York Times*, May 10, 2023, <https://www.nytimes.com/2023/05/08/business/energy-environment/norway-electric-vehicles.html>, accessed August 2023.
- ⁹³ Government of the United Kingdom, "Government Takes Historic Step towards Net-Zero with End of Sale of New Petrol and Diesel Cars by 2030," November 18, 2020, <https://tinyurl.com/3492nnt7>, accessed August 2023.
- ⁹⁴ Kate Abnett, "EU Lawmakers Approve Effective 2035 Ban on New Fossil Fuel Cars," Reuters, February 13, 2023, <https://tinyurl.com/bdeukbx5>, accessed August 2023.
- ⁹⁵ Alexandru Arba, "Sales Share of Electric Vehicles in Japan from 2013 to 2022," Statista, June 7, 2023, <https://www-statista-com.ezp-prod1.hul.harvard.edu/statistics/711994/japan-electric-car-market-share/>, accessed August 2023.
- ⁹⁶ Ewing, "In Norway, the Electric Vehicle Future Has Already Arrived."
- ⁹⁷ Hawkins, "Battery Power: How China Could Take Control of the Electric Vehicle Market."
- ⁹⁸ Abnett, "EU Lawmakers Approve Effective 2035 Ban on New Fossil Fuel Cars."
- ⁹⁹ Tom Krisher and Matthew Daly, "AP Sources: EPA Car Rule to Push Huge Increase in EV Sales," Associated Press, April 10, 2023, <https://tinyurl.com/yxup3hww>, accessed July 2023.
- ¹⁰⁰ Krisher and Daly, "AP Sources: EPA Car Rule to Push Huge Increase in EV Sales."
- ¹⁰¹ "Electric Vehicles in South Korea," via Statista, June 8, 2023, <https://tinyurl.com/y8ceyayc>, accessed August 2023.
- ¹⁰² U.S. International Trade Association, "Japan Transition to Electric Vehicles," July 7, 2021, <https://www.trade.gov/market-intelligence/japan-transition-electric-vehicles>, accessed August 2023.
- ¹⁰³ Sheila Chiang, "Hyundai Explains How It Plans to Become One of the World's Top Three EV Makers by 2030," CNBC, May 10, 2023, <https://tinyurl.com/98u7wek9>; "Factbox: How U.S. Electric Vehicle Subsidy Rules Impact Europe," Reuters, March 20, 2023, <https://tinyurl.com/h8ybuufy>; both accessed August 2023.
- ¹⁰⁴ Reuters, "Germany to Reduce Electric Car Subsidies in 2023," July 26, 2022, <https://tinyurl.com/2ab2t4cw>, accessed August 2023.
- ¹⁰⁵ Reuters, "Germany to Reduce Electric Car Subsidies in 2023."
- ¹⁰⁶ White, "How China Cornered the Market for Clean Tech."

- ¹⁰⁷ Center for Strategic and International Studies estimates, cited in Yoku Kubota and Selina Cheng, "In China, the Era of Western Carmakers Is Over," July 10, 2023, <https://tinyurl.com/5beu8vm4>, accessed July 2023.
- ¹⁰⁸ Gerard DiPippo, Ilaria Mazzocco, and Scott Kennedy, "Red Ink: Estimating Chinese Industrial Policy Spending in Comparative Perspective," *The Center for Strategic and International Studies*, May 2022, page 56, <https://tinyurl.com/57wz745j>, accessed August 2023.
- ¹⁰⁹ DiPippo, Mazzocco, and Kennedy, "Red Ink: Estimating Chinese Industrial Policy Spending in Comparative Perspective," page 56.
- ¹¹⁰ Hinata, "BYD Eyes Overtaking Tesla This Year as World's Top EV Maker."
- ¹¹¹ Reuters, "Tesla, EV Rivals Absorb Costs after China Pulls Plug on Subsidy," *Reuters*, January 5, 2023, <https://tinyurl.com/3jdmstwa>, accessed July 2023.
- ¹¹² Qiaoyi Li and Liz Lee, "China Unveils \$72 Billion Tax Break for EVs, Other Green Cars to Spur Demand," *Reuters*, June 21, 2023, <https://tinyurl.com/2d97eytt>, accessed August 2023.
- ¹¹³ Sasha Lekach and Alex Kwanten, "EV Survey Shows Range Anxiety High While Charging Costs Remain Low," *Forbes*, June 28, 2022, <https://www.forbes.com/wheels/features/ev-range-cost-confidence-survey/>, accessed August 2023.
- ¹¹⁴ David Ferris, "Why America's EV chargers Keep Breaking," *Politico*, April 12, 2023, <https://tinyurl.com/mwrtsc22>, accessed August 2023.
- ¹¹⁵ Ben Ward, "China's EV makers rev up, surpassing expectations as Russian demand drives gains," *NASDAQ*, July 4, 2023, <https://tinyurl.com/4xewy2t6>, accessed July 2023.
- ¹¹⁶ "Hyundai Motor Tries U-Turn in China Strategy," *Korea JoongAng Daily*, January 21, 2022, <https://tinyurl.com/y82xhwca>, accessed August 2023.
- ¹¹⁷ IEA "Global EV Outlook 2023," <https://tinyurl.com/4w4bb9sv>, accessed August 2023.
- ¹¹⁸ Mak, "Chinese NEV Makers Need to Offer Better Value Proposition to Buyers as Price War 'Unsustainable in the Long-Term', Says Fitch."
- ¹¹⁹ Mak, "Chinese NEV Makers Need to Offer Better Value Proposition to Buyers as Price War 'Unsustainable in the Long-Term', Says Fitch."
- ¹²⁰ Daniel Ren, "Tesla Might Lose Ground to Local Pretenders, as Likes of Xpeng, BYD Steal a March with Made-for-China EVs," *South China Morning Post*, April 8, 2023, <https://tinyurl.com/4ht9pu84>, accessed July 2023.
- ¹²¹ Ren, "Tesla Might Lose Ground to Local Pretenders, as Likes of Xpeng, BYD Steal a March with Made-for-China EVs."
- ¹²² Ren, "Tesla Might Lose Ground to Local Pretenders, as Likes of Xpeng, BYD Steal a March with Made-for-China EVs."
- ¹²³ Ren, "Tesla Might Lose Ground to Local Pretenders, as Likes of Xpeng, BYD Steal a March with Made-for-China EVs."
- ¹²⁴ Lai, "China's Electric Vehicle Market, a Rising Global Leader in EV Technology."
- ¹²⁵ Ren, "Tesla Might Lose Ground to Local Pretenders, as Likes of Xpeng, BYD Steal a March with Made-for-China EVs."
- ¹²⁶ Reuters, "Battle for China's Electric SUV Market Heats Up at Home and Abroad."
- ¹²⁷ Eddy, "China's Leading Electric Carmaker Has Arrived in Germany."
- ¹²⁸ "Electric Vehicles in South Korea," via Statista, June 8, 2023, <https://tinyurl.com/y8ceyayc>, accessed August 2023.
- ¹²⁹ "Electric Vehicles in South Korea," via Statista, June 8, 2023, <https://tinyurl.com/y8ceyayc>, accessed August 2023.
- ¹³⁰ "Electric Vehicles in South Korea," via Statista, June 8, 2023, <https://tinyurl.com/y8ceyayc>, accessed August 2023.
- ¹³¹ Krisher and Daly, "AP Sources: EPA Car Rule to Push Huge Increase in EV Sales."
- ¹³² Alison Spencer, Stephanie Ross, and Alec Tyson, "How Americans View Electric Vehicles," *Pew Research Center*, July 13, 2023, <https://tinyurl.com/bddrk8n2>, accessed July 2023.
- ¹³³ Spencer, Ross, and Tyson, "How Americans View Electric Vehicles."

- ¹³⁴ Spencer, Ross, and Tyson, "How Americans View Electric Vehicles."
- ¹³⁵ Spencer, Ross, and Tyson, "How Americans View Electric Vehicles."
- ¹³⁶ Medora Lee, "Why Are Americans Less Interested in Owning an EV? Cost and Charging Still Play a Part," *USA Today*, July 27, 2023, <https://tinyurl.com/ydvntpha>, accessed July 2023.
- ¹³⁷ Spencer, Ross, and Tyson, "How Americans View Electric Vehicles."
- ¹³⁸ Mathilde Carlier, "Europe: EV purchase incentives in selected countries by type of incentive 2023," Statista, March 31, 2023, <https://tinyurl.com/25vbw2xy>, accessed July 2023.
- ¹³⁹ "Electric Vehicles," via Statista Outlook, May 2023, <https://tinyurl.com/49c94bby>, accessed August 2023.
- ¹⁴⁰ Horst Schneider, Niccolo Serra, John Murphy, Ming Hsun Lee, Joon-Ho Lee, Kei Nihonyanagi, K. J. Hwang, and Michael Jacks, "EV Tracker Apr-23: EU Loses Significant Share; Special Analysis on Premium EVs," Bank of America Global Research, May 30, 2023, p. 2, via Refinitiv Workspace, accessed July 2023.
- ¹⁴¹ Reuters, "Battle for China's Electric SUV Market Heats Up at Home and Abroad."
- ¹⁴² Eddy, "China's Leading Electric Carmaker Has Arrived in Germany."
- ¹⁴³ Nick Carey, "China Poses Greatest Risk to Europe's Carmakers – Allianz Trade Study," Reuters, May 8, 2023, <https://tinyurl.com/mr2mkvd3>, accessed July 2023.
- ¹⁴⁴ Cardie Thompson, "Tesla history's most important moments, from its founders' launch to bringing EVs mainstream," *Business Insider*, June 22, 2023, <https://www.businessinsider.com/tesla-history-founders#2021-19>, accessed August 2023.
- ¹⁴⁵ Eddy, "China's Leading Electric Carmaker Has Arrived in Germany."
- ¹⁴⁶ Eddy, "China's Leading Electric Carmaker Has Arrived in Germany."
- ¹⁴⁷ Eddy, "China's Leading Electric Carmaker Has Arrived in Germany."
- ¹⁴⁸ Eddy, "China's Leading Electric Carmaker Has Arrived in Germany."
- ¹⁴⁹ Eddy, "China's Leading Electric Carmaker Has Arrived in Germany."
- ¹⁵⁰ Sky Ariella, "The 11 Largest Car Companies in the World," Zippia, April 13, 2023, <https://tinyurl.com/27z5tb5c>, accessed August 2023.
- ¹⁵¹ Schneider et al, "EV Tracker Apr-23," Bank of America Global Research, p. 10.
- ¹⁵² Abnett, "EU Lawmakers Approve Effective 2035 Ban on New Fossil Fuel Cars."
- ¹⁵³ William Milkes, Monica Raymunt, and Stefan Nicola, "VW and Mercedes's Electric-Car Ambitions Run into Trouble," Bloomberg, November 19, 2022, <https://tinyurl.com/59am2k8f>, accessed August 2023.
- ¹⁵⁴ Schneider et al, "EV Tracker Apr-23," Bank of America Global Research, p. 10.
- ¹⁵⁵ Counterpoint Research, "Global Passenger Electric Vehicle Market Share," June 13, 2023, <https://www.counterpointresearch.com/global-electric-vehicle-market-share/>, accessed August 2023.
- ¹⁵⁶ Milkes, Raymunt, and Nicola, "VW and Mercedes's Electric-Car Ambitions Run into Trouble."
- ¹⁵⁷ Milkes, Raymunt, and Nicola, "VW and Mercedes's Electric-Car Ambitions Run into Trouble."
- ¹⁵⁸ Milkes, Raymunt, and Nicola, "VW and Mercedes's Electric-Car Ambitions Run into Trouble."
- ¹⁵⁹ Milkes, Raymunt, and Nicola, "VW and Mercedes's Electric-Car Ambitions Run into Trouble."
- ¹⁶⁰ Jacky Wong, "Volkswagen's EV Gambit in China Makes Sense," *The Wall Street Journal*, July 27, 2023, <https://www.wsj.com/articles/volkswagens-ev-gambit-in-china-makes-sense-acc68f54>, accessed August 2023.
- ¹⁶¹ Wong, "Volkswagen's EV Gambit in China Makes Sense."
- ¹⁶² Wong, "Volkswagen's EV Gambit in China Makes Sense."
- ¹⁶³ Schneider et al, "EV Tracker Apr-23," Bank of America Global Research, p. 10.

- ¹⁶⁴ Milkes, Raymunt, and Nicola, "VW and Mercedes's Electric-Car Ambitions Run into Trouble."
- ¹⁶⁵ Nick Gibbs, "BMW Expects New EVs to Match Combustion-Engine Cars on Profit," *Automotive News Europe*, May 12, 2022, <https://tinyurl.com/yvj33dxw>, accessed August 2023.
- ¹⁶⁶ Victoria Waldersee, "BMW cautiously optimistic on 2023, EV targets ahead of schedule," *Reuters*, March 15, 2023, <https://tinyurl.com/tp9xtbkx>, accessed August 2023.
- ¹⁶⁷ Waldersee, "BMW cautiously optimistic on 2023, EV targets ahead of schedule."
- ¹⁶⁸ Shirley Lu, George Serafeim, and Michael W. Toffel, "Driving Decarbonization at BMW," HBS 123-008 (Boston: Harvard Business Publishing, 2022), p. 2.
- ¹⁶⁹ Schneider et al, "EV Tracker Apr-23," Bank of America Global Research, p. 2.
- ¹⁷⁰ Fareed Zakaria, "The rest of the world doesn't see China the way we do," *The Washington Post*, June 9, 2023, <https://www.washingtonpost.com/opinions/2023/06/09/europe-asia-china-policy-trade/>, accessed August 2023.
- ¹⁷¹ Zakaria, "The rest of the world doesn't see China the way we do."
- ¹⁷² "China's NIO takes on BMW, Mercedes in home market with flexible battery plan," *Reuters*, September 21, 2022, <https://tinyurl.com/3usmdhr7>, accessed August 2023.
- ¹⁷³ Isaac Atienza, "This Electric Wagon from Nio Has Beaten BMW to the Market," *Top Speed*, June 22, 2023, <https://www.topspeed.com/nio-electric-wagon-beats-bmw-to-the-market/>, accessed August 2023.
- ¹⁷⁴ Atienza, "This Electric Wagon from Nio Has Beaten BMW to the Market."
- ¹⁷⁵ "China's NIO takes on BMW, Mercedes in home market with flexible battery plan," *Reuters*, September 21, 2022, <https://tinyurl.com/3usmdhr7>, accessed August 2023.
- ¹⁷⁶ Schneider et al, "EV Tracker Apr-23," Bank of America Global Research, p. 2.
- ¹⁷⁷ Bloomberg, "BMW i3 Is a Cult Classic in U.S.," December 13, 2022, available at *Automotive News Europe*, <https://europe.autonews.com/automakers/bmw-i3-ev-cult-classic-us>, accessed August 2023.
- ¹⁷⁸ Bloomberg, "BMW i3 Is a Cult Classic in U.S."
- ¹⁷⁹ Gibbs, "BMW expects new EVs to match combustion-engine cars on profit," *Automotive News Europe*, May 12, 2022, <https://tinyurl.com/5n7u9njx>, accessed August 2023.
- ¹⁸⁰ Stephan Thomke, "BMW AG: The Digital Car Project," HBS No. 699-044, (Boston: Harvard Business School Publishing, 2001), p. 1.
- ¹⁸¹ Ashwini Choudhary, "The Convergence of Technology and Manufacturing in the Automotive Sector," *Forbes*, June 19, 2020, <https://tinyurl.com/enytevx6>, and Terry Shea, "Why Does It Cost So Much for Automakers to Develop New Models?" *Autoblog*, July 27, 2010, <https://tinyurl.com/29em3zzf>, both accessed August 2023.
- ¹⁸² Gibbs, "BMW Expects New EVs to Match Combustion-Engine Cars on Profit."
- ¹⁸³ Gibbs, "BMW Expects New EVs to Match Combustion-Engine Cars on Profit," Performance and cost information: Tina Casey, "BMW's Excellent Solid-State EV Battery Adventure," *CleanTechnica*, January 21, 2023, <https://tinyurl.com/4rvespkd>; both accessed August 2023.
- ¹⁸⁴ Peter Johnson, "BMW Adds EV Battery Capacity in Germany to Ramp Up Electric Mini Countryman Production," *Electrek*, October 21, 2022, <https://tinyurl.com/y6ff46ch>, accessed August 2023.
- ¹⁸⁵ Schneider et al, "EV Tracker Apr-23," Bank of America Global Research, p. 2.
- ¹⁸⁶ Schneider et al, "EV Tracker Apr-23," Bank of America Global Research, p. 2.
- ¹⁸⁷ Russ Mitchell, "Forget Tesla. Mercedes is betting \$45 billion it can become the king of luxury EVs," *Los Angeles Times*, July 20, 2023, <https://tinyurl.com/22jih6bt>, accessed August 2023.
- ¹⁸⁸ "BYD's Welcome in Europe Worries Local Automakers," *Automotive News Europe*, October 22, 2022, <https://europe.autonews.com/paris-auto-show/byds-welcome-europe-worries-local-automakers>, accessed July 2023.

- ¹⁸⁹ Anmar Frangoul, "A Marathon, Not a Sprint: Mercedes-Benz CEO Outlines Challenges in Shift to EVs," CNBC, July 28, 2023, <https://tinyurl.com/4fba7tpp>, accessed August 2023.
- ¹⁹⁰ Martin Bigg, "A Catastrophic Snag," CarBuzz, March 13, 2021, <https://tinyurl.com/4y5yeprij>, accessed August 2023.
- ¹⁹¹ Bigg, "A Catastrophic Snag."
- ¹⁹² Milkes, Raymunt, and Nicola, "VW and Mercedes's Electric-Car Ambitions Run into Trouble."
- ¹⁹³ Milkes, Raymunt, and Nicola, "VW and Mercedes's Electric-Car Ambitions Run into Trouble."
- ¹⁹⁴ Andrew J. Hawkins, "Mercedes-Benz is the first to bring Level 3 automated driving to the US," *The Verge*, January 27, 2023, <https://tinyurl.com/y5sthw2t>; Peter Sigal, "Mercedes opens sales of Level 3 self-driving system on S-Class, EQS," *Automotive News Europe*, May 6, 2022, <https://tinyurl.com/8v8e75z8>, both accessed August 2023.
- ¹⁹⁵ Mike Costello, "Toyota Remains World's Biggest Car Company," CarExpert, January 30, 2023, <https://tinyurl.com/bdhd53p>, accessed August 2023.
- ¹⁹⁶ Costello, "Toyota Remains World's Biggest Car Company."
- ¹⁹⁷ Benson Ojango, "The Real Reason Toyota Shuns Electric Cars," Top Speed, April 5, 2023, <https://tinyurl.com/6dx8u7yj>, accessed August 2023.
- ¹⁹⁸ Steve Levine, "Is Toyota Right to Be Skeptical about EVs?" The Information, August 14, 2023, <https://tinyurl.com/2b4k6wum>, accessed August 2023.
- ¹⁹⁹ Alina Moore, "10 Gamechanging Hybrid Vehicles that Hit the Market," Top Speed, July 10, 2023, <https://tinyurl.com/2x6ca3rr>, accessed August 2023.
- ²⁰⁰ River Davis, "New Toyota Chief Says He Will Accelerate Shift in EV Strategy," *The Wall Street Journal*, February 13, 2023, <https://tinyurl.com/bdzejv4t>, accessed August 2023.
- ²⁰¹ Davis, "New Toyota Chief Says He Will Accelerate Shift in EV Strategy."
- ²⁰² James O'Neill, "With the 745-mile Solid-State Battery, Toyota Just Became a Force to Reckon With," Top Speed, July 22, 2023, <https://www.topspeed.com/toyota-745-mile-solid-state-battery/>, accessed August 2023.
- ²⁰³ O'Neill, "With the 745-mile Solid-State Battery, Toyota Just Became a Force to Reckon With."
- ²⁰⁴ Mary Swansburg, "Toyota's Newest Breakthrough Could Be the 'Kiss of Death' for Gas-Powered Cars — And Could Hit the Market as Early as 2027," Yahoo! News, August 13, 2023, <https://tinyurl.com/3fer3wws>, accessed August 2023.
- ²⁰⁵ Peter Johnson, "Toyota Falters as Japan's EV Hesitancy Claims Second Victim in China," Electrek, July 24, 2023, <https://electrek.co/2023/07/24/toyota-falters-japans-ev-hesitancy-takes-toll-china/>, accessed August 2023.
- ²⁰⁶ Johnson, "Toyota Falters as Japan's EV Hesitancy Claims Second Victim in China."
- ²⁰⁷ Johnson, "Toyota Falters as Japan's EV Hesitancy Claims Second Victim in China."
- ²⁰⁸ Johnson, "Toyota Falters as Japan's EV Hesitancy Claims Second Victim in China."
- ²⁰⁹ "Electric Vehicles in South Korea," via Statista, June 8, 2023, <https://tinyurl.com/y8ceyayc>, accessed August 2023.
- ²¹⁰ "Electric Vehicles in South Korea," via Statista.
- ²¹¹ Chiang, "Hyundai Explains How It Plans to Become One of the World's Top Three EV Makers by 2030."
- ²¹² Chiang, "Hyundai Explains How It Plans to Become One of the World's Top Three EV Makers by 2030."
- ²¹³ Chiang, "Hyundai Explains How It Plans to Become One of the World's Top Three EV Makers by 2030."
- ²¹⁴ Chiang, "Hyundai Explains How It Plans to Become One of the World's Top Three EV Makers by 2030."
- ²¹⁵ Nan-Sae Bin and Il-Gue Kim, "Hyundai Motor Breaks Out of Three-Year Slump in China," *The Korean Economic Daily*, July 9, 2023, <https://www.kedglobal.com/automobiles/newsView/ked202307090001>, accessed August 2023.
- ²¹⁶ Jeff Amy, "Biden's Inflation Reduction Act Is Bringing an Up to \$5 Billion Electric Battery Plant to the Atlanta Suburbs," *Fortune*, December 8, 2022, <https://tinyurl.com/s6upmvse>, accessed August 2023.

- ²¹⁷ Amy, "Biden's Inflation Reduction Act Is Bringing an up to \$5 Billion Electric Battery Plant to the Atlanta Suburbs."
- ²¹⁸ Amy, "Biden's Inflation Reduction Act is bringing an up to \$5 billion electric battery plant to the Atlanta suburbs."
- ²¹⁹ Thompson, "Tesla History's Most Important Moments, from Its Founders' Launch to Bringing EVs Mainstream."
- ²²⁰ Thompson, "Tesla History's Most Important Moments, from Its Founders' Launch to Bringing EVs Mainstream."
- ²²¹ Thompson, "Tesla History's Most Important Moments, from Its Founders' Launch to Bringing EVs Mainstream."
- ²²² Tim Levin, "Tesla Is Still the King of EVs – But Rivals Will Catch Up Fast, Wall Street Analysts Say," *Business Insider*, June 21, 2021, <https://tinyurl.com/bdfx8s3y>, accessed August 2023.
- ²²³ Levin, "Tesla Is Still the King of EVs – But Rivals Will Catch Up Fast, Wall Street Analysts Say."
- ²²⁴ "Electric Vehicles, A Global Overview," via Statista, <https://tinyurl.com/mr44ystk>, accessed August 2023.
- ²²⁵ David Golman, "Tesla delivered a record 1.3 million vehicles in 2022, but it still disappointed Wall Street," *CNN*, January 3, 2023, <https://tinyurl.com/6pubzm9a>, accessed August 2023.
- ²²⁶ Reuters, "Tesla Shares Continue Slide with Shrinking Demand and Logistics Snags," *The Guardian*, January 3, 2023, <https://www.theguardian.com/technology/2023/jan/03/tesla-shares-price-value-decline-elon-musk>, accessed August 2023.
- ²²⁷ Reuters, "Tesla Shares Continue Slide with Shrinking Demand and Logistics Snags," Max Zahn, "A Timeline of Elon Musk's Tumultuous Twitter Acquisition," *ABC News*, November 11, 2022, <https://tinyurl.com/cf7eed5k>, accessed August 2023.
- ²²⁸ Suvrat Kothari, "Tesla Sales Soar, but Analysts Forecast Challenging Business Conditions," *InsideEVs*, July 3, 2023, <https://insideevs.com/news/674938/bernstein-analyst-predicts-business-challenges-for-tesla/>, accessed August 2023.
- ²²⁹ Kothari, "Tesla to Lose Significant EV Market Share to Ford, GM by 2026: Report."
- ²³⁰ Rebecca Elliott, "Tesla Deliveries Rose 83% in the Latest Quarter," *Wall Street Journal*, July 2, 2023, <https://www.wsj.com/articles/tesla-deliveries-to-show-whether-price-cuts-are-paying-off-2f2aac93>, accessed August 2023.
- ²³¹ Al Root, "Tesla, Ford, and the EV Inventory Debate on Wall Street," *Barron's*, July 19, 2023, <https://www.barrons.com/articles/tesla-ev-stock-ford-9a108795>, accessed August 2023.
- ²³² Root, "Tesla, Ford, and the EV Inventory Debate on Wall Street."
- ²³³ Joseph White, "Ford Revs Up Electric Truck Output, Testing Mainstream EV Demand," *Reuters*, August 1, 2023, <https://tinyurl.com/4mrwb9db>, accessed August 2023.
- ²³⁴ Root, "Tesla, Ford, and the EV Inventory Debate on Wall Street."
- ²³⁵ Thompson, "Tesla History's Most Important Moments, from Its Founders' Launch to Bringing EVs Mainstream."
- ²³⁶ Root, "Tesla, Ford, and the EV Inventory Debate on Wall Street."
- ²³⁷ Root, "Tesla, Ford, and the EV Inventory Debate on Wall Street."
- ²³⁸ Tesla, "Supercharger," <https://www.tesla.com/supercharger>, accessed August 2023.
- ²³⁹ Tesla, "Support: Charging Your Tesla," 2023, <https://www.tesla.com/support/charging>, accessed August 2023.
- ²⁴⁰ Jarrett Renshaw and Hyunjoon Jin, "New Biden EV Charger Rules Stress Made in America, Force Tesla Changes," *Reuters*, February 15, 2023, <https://tinyurl.com/522fsthu>, accessed August 2023.
- ²⁴¹ Renshaw and Jin, "New Biden EV Charger Rules Stress Made in America, Force Tesla Changes."
- ²⁴² Aditya Soni and Hyunjoon Jin, "Tesla jumps as GM deal makes its charging network closer to US standard," *Reuters*, June 9, 2023, <https://tinyurl.com/57takc2f>, accessed August 2023.
- ²⁴³ Adam Rogers, "The Real Reason Tesla Is Letting Rival Car Companies Use Its EV Charging Stations," *Business Insider*, July 6, 2023, tinyurl.com/msu7mhw5, accessed August 2023.
- ²⁴⁴ Dan Mihalascu, "Future Ford EVs 'Deep into Development,' Will Use Smaller Batteries," *InsideEVs*, February 3, 2023, <https://tinyurl.com/mr4r73z7>, accessed August 2023.

- ²⁴⁵ Alisa Priddle, "Codename Project T3: Ford Has Another New EV Pickup Coming in 2025," *Motor Trend*, March 24, 2023, <https://tinyurl.com/4hfuub39>, accessed August 2023.
- ²⁴⁶ Mihalascu, "Future Ford EVs 'Deep into Development,' Will Use Smaller Batteries."
- ²⁴⁷ Kevin Hunt, "The Ford F-150 Lightning Can Be Used as a Backup Generator with This \$3,895 System," *TechHive*, May 3, 2022, <https://tinyurl.com/kxebzy72>, accessed August 2023.
- ²⁴⁸ Mihalascu, "Future Ford EVs 'Deep into Development,' Will Use Smaller Batteries."
- ²⁴⁹ Mihalascu, "Future Ford EVs 'Deep into Development,' Will Use Smaller Batteries."
- ²⁵⁰ Mihalascu, "Future Ford EVs 'Deep into Development,' Will Use Smaller Batteries."
- ²⁵¹ Mihalascu, "Future Ford EVs 'Deep into Development,' Will Use Smaller Batteries."
- ²⁵² Joseph White and Nathan Gomes, "Ford's Profit Rises on Strong Truck Demand but EV Outlook Weighs," *Reuters*, May 2, 2023, <https://tinyurl.com/4yaretzz>, accessed August 2023.
- ²⁵³ Mike Colias, "Ford Says It Will Lose \$3 Billion on EVs This Year as It Touts Startup Mentality," *The Wall Street Journal*, March 23, 2023, <https://tinyurl.com/3fm9pyyj>, accessed August 2023.
- ²⁵⁴ Jackie Northam, "Ford Plans to Make EV Batteries in U.S. with Chinese Company That Developed the Tech," *NPR*, August 8, 2023, <https://tinyurl.com/uewwakjj>, accessed August 2023.
- ²⁵⁵ Northam, "Ford Plans to Make EV Batteries in U.S. with Chinese Company That Developed the Tech."
- ²⁵⁶ Northam, "Ford Plans to Make EV Batteries in U.S. with Chinese Company That Developed the Tech."
- ²⁵⁷ Northam, "Ford Plans to Make EV Batteries in U.S. with Chinese Company That Developed the Tech."
- ²⁵⁸ Jamie L. Lareau, "GM Reports Q1 Sales Gain with Record EV Sales, but Not of New Models," *Detroit Free Press*, April 3, 2023, <https://tinyurl.com/39ksy7w8>, accessed August 2023.
- ²⁵⁹ Claudia Assis, "GM's Stock Rises after Automaker Posts 19% Quarterly Sales Increase," *MarketWatch*, July 6, 2023, <https://tinyurl.com/muswfyzk>, accessed August 2023.
- ²⁶⁰ Michael Wayland, "EV Sales: Hyundai Overtakes GM, but Tesla's U.S. Dominance Continues," *CNBC*, July 7, 2023, <https://tinyurl.com/34zxna8>, accessed August 2023.
- ²⁶¹ Suvrat Kothari, "Tesla to Lose Significant EV Market Share to Ford, GM by 2026: Report," *Inside EVs*, July 3, 2023, <https://insideevs.com/news/674963/bank-of-america-car-wars-report/>, accessed August 2023.
- ²⁶² Kothari, "Tesla to Lose Significant EV Market Share to Ford, GM by 2026: Report."
- ²⁶³ Jameson Dow, "Cadillac Teases Electric Escalade IQ ahead of Reveal (Video)," *Electrek*, June 20, 2023, <https://tinyurl.com/yjcs2zrb>, accessed August 2023.
- ²⁶⁴ Wayland, "EV Sales: Hyundai Overtakes GM, but Tesla's U.S. Dominance Continues."
- ²⁶⁵ Kothari, "Tesla to Lose Significant EV Market Share to Ford, GM by 2026: Report."
- ²⁶⁶ Lareau, "GM Reports Q1 Sales Gain with Record EV Sales, but Not of New Models."
- ²⁶⁷ Mark Kane, "US: Chevrolet Bolt EV/Bolt EUV Sales Reached New Record In Q4 2022," *InsideEVs*, January 4, 2023, <https://insideevs.com/news/629487/us-chevrolet-bolt-ev-euv-sales-2022q4/>, accessed August 2023.
- ²⁶⁸ "All Electric 2023 Bolt EV," *GM*, <https://www.chevrolet.com/electric/bolt-ev>, accessed August 2023.
- ²⁶⁹ Scooter Doll, "GM Recalls 140,000 Chevy Bolt EVs Due to Risk of Carpet Catching Fire," *Electrek*, December 21, 2022, <https://tinyurl.com/5fnxash8>, accessed August 2023.
- ²⁷⁰ Dan Mihalascu, "GM Said to Extend 2023 Chevy Bolt EV/EUV Production into December," *InsideEVs*, August 12, 2023, <https://tinyurl.com/y7fp89tv>, accessed August 2023.
- ²⁷¹ Wayland, "EV sales: Hyundai Overtakes GM, but Tesla's U.S. Dominance Continues."

²⁷² Bailey Shultz, "Chevrolet Bolt Won't Be Retired after All. GM Says Nameplate Will Live On." *USA Today*, July 25, 2023, <https://tinyurl.com/mr3np3ca>, accessed August 2023.

²⁷³ Jess Weatherbed, "GM Is Making a New Chevy Bolt EV with an Ultium Battery," *The Verge*, July 25, 2023, <https://www.theverge.com/2023/7/25/23806863/chevy-bolt-ev-gm-ultium-battery>, accessed August 2023.

²⁷⁴ Weatherbed, "GM Is Making a New Chevy Bolt EV with an Ultium Battery."

²⁷⁵ Abdullah Tariq, "Everything We Know about GM's Ultium Battery Technology," *Top Speed*, August 2, 2023, <https://www.topspeed.com/everything-to-know-gm-ultium-battery-technology/>, accessed August 2023.

²⁷⁶ Weatherbed, "GM Is Making a New Chevy Bolt EV with an Ultium Battery."

²⁷⁷ "Electric Vehicles, A Global Overview," via Statista, <https://tinyurl.com/mr44ystk>, accessed August 2023.

²⁷⁸ Bhagchandani, "Forget Elon Musk, Meet Wang Chuanfu: the Chinese Billionaire Has Dethroned Tesla with BYD's 2022 Electric Car Sales and Was Backed by Warren Buffet—but Comes from a Family of Farmers."